

ALETHIA: closed-form in-context inference of dimension-six SMEFT effects in Drell-Yan

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Abstract

We ask whether drift monitoring can improve simulation-based inference in a physics foundation model when the model is allowed to acquire its own simulator queries through active learning. The vehicle is a $\mathcal{C}$ -agnostic in-context regressor for dimension-six Standard Model Effective Field Theory (SMEFT) effects in neutral-current Drell-Yan. The architecture is a closed-form linear-attention head whose forward pass is a single ridge solve in a learned feature map of the kinematic abscissa, with no gradient-descent step at inference. The closed form supplies four analytic properties used by the loop: the design-matrix leverage gives the predictive variance; the Cauchy-Schwarz inequality on the $\mathbf{A}^{-1}$ inner product gives a non-negative expected predictive information gain; the Sherman-Morrison identity updates $\mathbf{A}^{-1}$ in $\mathcal{O}(D^2)$ floating-point operations per added observation; the Eckart-Young inequality bounds the variance-amplification factor along the worst-conditioned direction by the condition number of the design matrix.

An augmented linear probe from $(\mathbf{w}_\theta, \log \mathbf{M}_{\text{ctx}})$ onto the Fisher-eigenbasis of the oracle reads out a per-direction Wilson posterior, evaluated identically on simulator contexts and admissible to a measured one. The reference benchmark is the bounded multi-restart maximum-likelihood estimator on the per-context analytic likelihood at $K = 12$, $\sigma_y = 0.05$. On the leading resolved direction the probe MSE is a factor of three below the MLE MSE in-box and a factor of five below on the withheld $|c_{\ell q}^{(3)}| \in [0.6, 1.0]$ band, with 68% interval coverage of 0.88 and 0.82 respectively. On the sub-leading direction the probe’s shrinkage adds more bias than it removes variance and the MLE is the tighter estimator; 68% coverage drops to 0.48 on both pools, and this is the direction the active-learning loop targets. On the two vertex directions, Fisher eigenvalues of $\sim 10^{-4}$ place the per-context MLE in a multimodal regime with MSE ~ 8 in $\tilde{\mathcal{C}}$ -units; the probe sits at the prior floor at MSE ≈ 0.18 . The MLE confirms the floor is the observable rather than the readout.

Three monitoring quantities (two residual-based drift detectors on standardised residuals and per-region conformal coverage, plus a design-matrix condition-number context-shape signal) aggregate into an acquisition trigger. On the engineered $|c_{\ell q}^{(3)}| \in [0.6, 1.0]$ band the loop drives the root-mean-square error on a 50-point target grid from 265 at the seed context to 2.65 after 500 oracle calls. On a 1D mass observable scored in cross-section-ratio root-mean-square error, the closed-form Expected Predictive Information Gain does not outperform uniform random selection inside the drifted region. The 1D-mass observable is near-isotropic in the inferred posterior, which is the regime the active-learning literature flags as adverse for greedy information gain. We extend the analytic oracle with the forward-backward asymmetry to break the vertex degeneracy, derive a closed-form parameter-space EPIG that drives the Wilson posterior entropy directly in the Fisher eigenbasis, and run the corrected test on the angular observable at a stressed budget across a 1800-chain design-only sweep and a 900-chain closed-loop sweep. Parameter-space EPIG contracts the resolved-subspace posterior $\Sigma_{\tilde{\mathcal{C}}_2}$ by 10–47 σ more than uniform random selection on every cell; the analytic-morphing maximum-likelihood estimator on the same direction $\tilde{\mathcal{C}}_2$ is worse than under random selection by 7–47 σ on every cell, with the divergence sharpening rather than closing as K and the loop depth grow. The mechanism is that the top-decile information-gain candidates are perfectly collinear in the $\mathbf{A}^{-1}$ metric on every pool tested: targeted acquisition concentrates

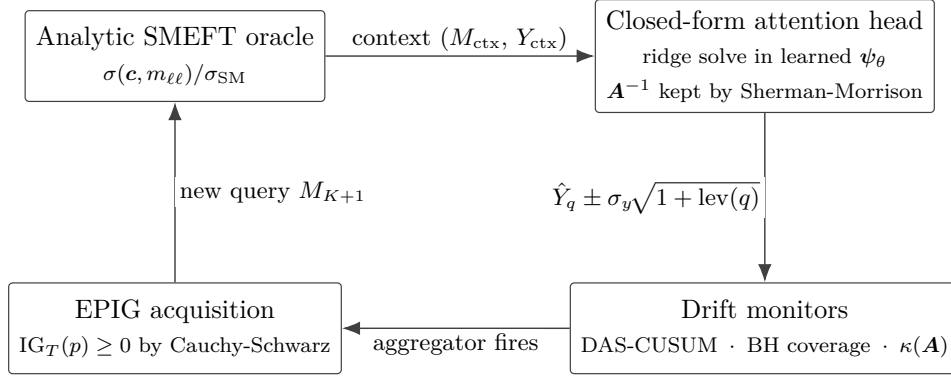


Figure 1: The ALETHIA closed loop. The analytic SMEFT oracle of Section 2 supplies a binned cross-section ratio at chosen $(\mathbf{c}, m_{\ell\ell})$. The closed-form attention head of Section 3 consumes the context $(M_{\text{ctx}}, Y_{\text{ctx}})$ through a ridge solve in a learned feature map ψ_θ and emits a prediction with closed-form predictive variance; the inverse design matrix $\mathbf{A}^{-1}$ is maintained by Sherman-Morrison rank-one updates at $\mathcal{O}(D^2)$ per added observation. Three drift monitors (a DAS-CUSUM on standardised residuals, a Benjamini-Hochberg-corrected per-region binomial coverage test, and an Eckart-Young condition-number monitor on the design matrix) drive an aggregator that escalates to EPIG acquisition, with Cauchy-Schwarz on the $\mathbf{A}^{-1}$ inner product rendering the information gain non-negative. The underlying ψ_θ weights are not touched at inference.

the posterior tightly along a one-dimensional ridge of the learned feature space that does not contain the bimodal target’s identifying combination, while random selection’s broader $m_{\ell\ell}$ coverage does. The active-learning component therefore yields a calibrated negative, paired with a closed-form contraction-MLE divergence signal that the dashboard exposes in real time.

The architectural comparison orders the closed-form Intention head, a fixed polynomial basis, a scaled-budget transformer JEPA, a Deep Sets pooling baseline, and a constraint-violating regressor by their M – Y alignment retention; the four closed-form properties used by the drift loop track that ordering. The closed-form head at 5 328 parameters fits the analytic oracle at median $R^2 = 0.9999$; the JEPA at 186 241 parameters and 10 000 scenarios reaches 0.95; Deep Sets at matched budget reaches 0.70 and discards the alignment that the closed-form ridge solve uses to read out Wilson directions.

The answer to the research question is structured. The SBI and drift components deliver: a calibrated Wilson posterior with under-coverage localised to one direction the Fisher eigenbasis identifies as the active-learning target, and a drift-triggered loop that closes mechanically on a controlled stress event. The active-learning component yields an information-geometry result that the multi-seed sweep makes explicit: closed-form parameter-space EPIG is at 10–47 σ ahead of uniform random selection on resolved-subspace posterior contraction across 2 700 chains, while the MLE estimator on the same direction is at 7–47 σ behind random selection across the same chains, with the divergence stable across K and loop depth. The closed-form contraction is therefore an over-confident proxy for inference quality on this physics, and the drift-monitored loop’s value is in surfacing the contraction-MLE divergence as a real-time signal, not in beating random with targeted acquisition.

1 Introduction

The research question is whether drift monitoring can be used to improve simulation-based inference in a physics foundation model through active learning. The components are independently established. Simulation-based inference (SBI) [1–3] amortises Bayesian inference over a prior of simulator outputs. Drift detection on a deployed predictive distribution is a standard streaming-monitoring task. Active learning selects the next training or context point from an acquisition functional. The system question is whether composing the three on the same closed-form inference primitive produces a Wilson posterior that the components in isolation do not, on

a controlled simulator where the truth is known. The vehicle is neutral-current Drell-Yan with dimension-six SMEFT effects, where the four-fermion operators $\mathcal{O}_{\ell q}^{(1,3)}$ and the vertex operators $\mathcal{O}_{Hq}^{(1,3)}$ together span the Fisher-eigenbasis split that makes the answer informative: two directions are resolved by the mass spectrum, two are not, and the loop has somewhere to do work.

The cluster of self-supervised foundation models in particle physics over the past two years, including masked-particle modelling [4], OmniJet- α [5], OmniLearn and OmniLearned [6, 7], HEP-JEPA [8], and Sophon [9], targets jet-level classification benchmarks measured by area under the receiver-operating-characteristic curve and background rejection at fixed signal efficiency. The contemporary SMEFT-side foundation model of Das Bakshi, Hobbs and Kriesten [10] trains a supervised contrastive embedding over 100 discrete Wilson configurations of neutral-current Drell-Yan and demonstrates Dirichlet-prior-network classification, anomaly detection, and nearest-neighbour retrieval on the embedding; the inference of Wilson coefficients from a measured distribution and the interplay with experimental data are left to future work. Neither cluster operates the foundation model as a drift-monitored, actively acquiring inference engine.

The architectural primitive used here is the closed-form linear-attention head of Garnelo and Czarnecki [11], in which a Bayesian linear regression on a learned feature map of the kinematic abscissa replaces the usual softmax-attention pattern. The closed form supplies four analytic properties on which the rest of the inference machinery depends.

The first is closed-form predictive variance. At a query point M_q , the predictive variance of the Bayesian linear regression on the learned feature map ψ_θ is $\sigma_y^2(1 + \text{lev}(q))$, with σ_y^2 the noise variance, the 1 the irreducible noise contribution at the query, and $\text{lev}(q) = \psi_\theta(M_q)^\top \mathbf{A}^{-1} \psi_\theta(M_q)$ the design-matrix leverage. The leverage is the Mahalanobis distance of the query in the inverse-design-matrix metric and equals the closed-form predictive variance of the corresponding Gaussian process under the kernel-ridge duality [12].

The second is closed-form acquisition. The expected predictive information gain of Smith et al. [13] on a target query T from a candidate pool point p is, in the same Bayesian linear regression, a closed-form function of lev_T , lev_p , and the cross-term $k_{Tp} = \psi_\theta(T)^\top \mathbf{A}^{-1} \psi_\theta(p)$. The Cauchy-Schwarz inequality on the $\mathbf{A}^{-1}$ inner product bounds $k_{Tp}^2 \leq \text{lev}_T \cdot \text{lev}_p$ and renders the information gain non-negative. The noise variance σ_y^2 cancels exactly under the homoscedastic assumption.

The third is closed-form sequential update. The inverse design matrix $\mathbf{A}^{-1}$ admits a rank-one update via the Sherman-Morrison identity [14] at $\mathcal{O}(D^2)$ floating-point operations per added observation. The context grows one observation at a time. The leverage is monotone non-increasing under context growth, which propagates to the target-set predictive entropy and gives a closed-form progress monitor for the loop.

The fourth is closed-form sensitivity. The condition number $\kappa(\mathbf{A})$ of the design matrix bounds the variance-amplification factor of the predictive variance along the worst-conditioned direction of feature space by the Eckart-Young theorem: a rank-one perturbation of $\mathbf{b}$ in the least-squares problem $\mathbf{A}\mathbf{w} = \mathbf{b}$ induces a relative perturbation in $\mathbf{w}$ of $\kappa(\mathbf{A})$ times the relative perturbation in $\mathbf{b}$. The condition number is therefore a coverage-drift signal that fires when new context observations explore previously unsampled directions of $\psi_\theta(M)$.

Together, properties (i)-(iv) close the inference loop. A leverage-stratified split-conformal calibrator [15–17] maintains finite-sample coverage of the predictive intervals against a stream of incoming residuals. A per-region binomial coverage test with the Benjamini-Hochberg [18] correction supplies one drift signal; the Data-Adaptive Symmetric CUSUM [19] on standardised residuals supplies a second; the condition-number monitor on the design matrix supplies a third. When any of the three signals fires, the closed-form information gain selects high-leverage simulator queries inside the drifted region, the Sherman-Morrison update incorporates the returned labels into the context, and the conformal calibrator refits.

The SBI deliverable is a per-direction Wilson posterior. The closed-form head produces an implicit ridge weight $\mathbf{w}_\theta$ per context; an augmented linear probe from $(\mathbf{w}_\theta, \log \mathbf{M}_{\text{ctx}})$ onto the

Fisher-eigenbasis of the oracle on the training Wilson prior reads out a Gaussian posterior in the rotated coordinates $\tilde{\mathbf{c}} = \mathbf{V}^\top \mathbf{c}$. Section 5 reports the readout against a bounded multi-restart maximum-likelihood estimator on the per-context analytic likelihood at $K = 12$ and $\sigma_y = 0.05$: on the leading resolved direction the probe dominates the MLE on both variance and bias, with probe MSE a factor of three below MLE MSE in-box and a factor of five on the withheld band; on the sub-leading direction the probe’s prior shrinkage adds more bias than it removes variance and the MLE is the tighter estimator, with under-coverage at the nominal 68% that the active-learning loop targets; on the vertex directions the per-context likelihood is multimodal at this K and the probe sits at the prior floor, where the MLE confirms the floor is the observable.

The closed-loop demonstration of Section 4 runs the drift-actively-acquire-recalibrate loop on the engineered $|c_{\ell q}^{(3)}| \in [0.6, 1.0]$ band. The setup is engineered: $|c| = 0.8$ at $\Lambda = 1$ TeV violates the EFT validity condition $\hat{s}/\Lambda^2 < 1$ above $m_{\ell\ell} \approx 1$ TeV, and the cross-section-ratio recovery is function recovery on a controlled out-of-distribution curve. Inside the controlled setting the loop drives the root-mean-square error on a 50-point target grid from 265 at the seed context of eight observations to 2.65 after 500 oracle calls. The recovery establishes that the loop closes on a designed-difficult drift event. The companion test of whether the active-learning backend itself contributes is harder. On the 1D mass observable scored in cross-section-ratio root-mean-square error, the EPIG acquisition functional is interchangeable with uniform random selection inside the drifted region; this is consistent with the Kirsch et al. analysis of batched information gain on near-isotropic targets [20]. Section 4.7 reports the diagnosis: the 1D mass observable is near-isotropic in the inferred Wilson posterior, so μ -space coverage on the target band cannot reward acquisition design. The right test rotates into the Fisher eigenbasis on an observable that breaks the vertex degeneracy. Section 2.1 extends the analytic oracle with the forward-backward asymmetry, validated against a Monte Carlo cross-check at the per-cent level; Section 4.8 derives a closed-form parameter-space EPIG that drives the Wilson posterior entropy in the eigenbasis; Section 4.9 reports the 2700-chain sweep that resolves the AL component as a contraction-MLE divergence that the closed-form ridge head exposes in real time.

Section 6 reports the architectural comparison on held-out scenarios drawn from the analytic SMEFT oracle. The principal contrast is between the learned and fixed-polynomial feature maps of the same closed-form ridge, which isolates what the learning contributes: the learned basis matches the median ($R^2 = 0.9999$ versus 0.99) and reduces fifth-percentile dispersion on tail scenarios. Deep Sets [4, 21] enters as an alignment ablation; pooling the context into a fixed-dimensional summary discards joint alignment, and median R^2 drops to 0.70. A scaled-budget transformer Joint-Embedding Predictive Architecture [22, 23] recovers comparable median R^2 at thirty times the parameter count, characterising the parameter-and-data cost of learning the alignment the closed form has built in. Section 6.6 orders all five architectures by their M - Y alignment retention and shows the four closed-form properties used by the drift loop track that ordering.

The closed-form head is $\mathbf{c}$ -agnostic by construction. Equation (3) below has no slot for a Wilson coefficient. The same forward pass accepts a binned cross-section ratio drawn from a simulator, a Monte Carlo replica, or a measured differential distribution; the inference machinery treats the Wilson coefficient as a latent quantity to be recovered from the implicit ridge weight. At 5328 trainable parameters the model is at the opposite end of the scale from contemporary self-supervised foundation models in particle physics. The foundation-model property in use here is the readiness of the same forward pass to consume an arbitrary context within the SMEFT prior, including a future measured one.

The paper is organised as follows. Section 2 fixes the SMEFT conventions in Drell-Yan, describes the oracles, and reports the angular-observable extension. Section 3 introduces the closed-form attention head. Section 4 reports the closed-loop system, the engineered drift demonstration, and the diagnosis of the EPIG-versus-random null on the 1D observable. Section 5 reports the Wilson posterior, the per-direction coverage and the MLE-efficiency analysis. Section 6

Table 1: Warsaw-basis dimension-six operators contributing to neutral-current Drell-Yan production $pp \rightarrow \mu^+ \mu^-$ at tree level, organised by the leading power counting of their interference with, or square against, the Standard Model amplitude. Here v is the electroweak scale and $\hat{s}$ the partonic centre-of-mass energy. The grouping follows [10] and the calculation of [25]. The four operators marked $\star$ span the directions used in the closed-loop demonstration of Section 4 and the empirical comparison of Section 6.

power counting	operators
$\mathcal{O}(\hat{s}/\Lambda^2)$	$\mathcal{O}_{\ell q}^{(1)\star}, \mathcal{O}_{\ell q}^{(3)\star}, \mathcal{O}_{\ell u}, \mathcal{O}_{\ell d}, \mathcal{O}_{eu}, \mathcal{O}_{ed}, \mathcal{O}_{qe}$
$\mathcal{O}(v^2/\Lambda^2)$	$\mathcal{O}_{H\ell}^{(1)}, \mathcal{O}_{H\ell}^{(3)}, \mathcal{O}_{He}, \mathcal{O}_{Hq}^{(1)\star}, \mathcal{O}_{Hq}^{(3)\star}, \mathcal{O}_{Hu}, \mathcal{O}_{Hd}, \mathcal{O}_{HD}, \mathcal{O}_{HWB}$
$\mathcal{O}(v^4/\Lambda^4)$	$\mathcal{O}_{H\Box}, \mathcal{O}_{e\varphi}, \mathcal{O}_{u\varphi}, \mathcal{O}_{d\varphi}$
$\mathcal{O}(v^2\hat{s}/\Lambda^4)$	$\mathcal{O}_{eW}, \mathcal{O}_{eB}, \mathcal{O}_{uW}, \mathcal{O}_{uB}, \mathcal{O}_{dW}, \mathcal{O}_{dB}, \mathcal{O}_{HG}$
$\mathcal{O}(\hat{s}^2/\Lambda^4)$	$\mathcal{O}_{\ell eqd}, \mathcal{O}_{\ell equ}^{(1)}, \mathcal{O}_{\ell equ}^{(3)}$

reports the architectural comparison and the alignment-retention table. Section 7 positions the work against the literature and discusses the measured-data extension; Section 8 closes. Appendix A collects closed-form derivations including the parameter-space EPIG, Appendix B reports a polynomial-toy verification, Appendix C reports representation-level diagnostics, and Appendix D gives reproduction details.

2 SMEFT in Drell-Yan and the oracle

The dimension-six SMEFT extension of the Standard Model is written as

$$\mathcal{L}_{\text{SMEFT}} = \mathcal{L}_{\text{SM}} + \sum_i \frac{c_i}{\Lambda^2} \mathcal{O}_i + \mathcal{O}(\Lambda^{-4}), \quad (1)$$

with $\{\mathcal{O}_i\}$ the operators of the Warsaw basis [24] and Λ the effective field theory scale. The Warsaw basis contains 2 499 baryon- and lepton-number-conserving Wilson coefficients at dimension six. Imposing a $U(3)^5$ flavour symmetry and restricting to real, CP-conserving coefficients reduces this to 60. For neutral-current Drell-Yan only a subset enters at tree level and at $\mathcal{O}(\Lambda^{-2})$, organised in Table 1 by the leading power counting of their contribution to the partonic cross section.

The differential cross section, integrated over angular variables and binned in the invariant dilepton mass $m_{\ell\ell}$, decomposes exactly as

$$\sigma(\mathbf{c}, m_{\ell\ell}) = \sigma_{\text{SM}}(m_{\ell\ell}) + \sum_i A_i(m_{\ell\ell}) c_i + \sum_{i \leq j} B_{ij}(m_{\ell\ell}) c_i c_j, \quad (2)$$

with the linear coefficients A_i from interference of the dimension-six amplitude with the Standard Model amplitude and the quadratic coefficients B_{ij} from the squared dimension-six amplitude. The four-fermion contributions grow as $(m_{\ell\ell}/\Lambda)^4$ in the high-mass tail, which produces the strongest experimental sensitivity at high invariant mass and motivates the choice of $m_{\ell\ell}$ as the kinematic axis throughout this work.

We use two oracles for the cross section as a function of $(\mathbf{c}, m_{\ell\ell})$. The first is an analytic leading-order calculation that implements equation (2) over the fourteen dimension-six operators of greatest phenomenological interest, convolved with the CT18NNLO parton distribution functions [26] via LHAPDF [27]. The second is a MadGraph5_aMC@NLO [28] pipeline that cross-validates the analytic oracle. The analytic and Monte Carlo cross sections agree at the per-cent level for $|\mathbf{c}| \leq 1$ across $m_{\ell\ell} \in [0.3, 2.5]$ TeV, with the disagreement growing with $|\mathbf{c}|$ as expected from the missing Λ^{-4} contributions in the analytic ansatz. The closed-loop run in Section 4 uses the analytic oracle; the Monte Carlo tier is exercised only as a validation gate on the angular-observable extension described next.

2.1 Angular-observable extension

The angle-integrated rate of equation (2) is diagonal in $\cos \theta^*$ and annihilates the antisymmetric forward-backward component, so the two vertex operators $c_{Hq}^{(1,3)}$ produce only rate-only modifications proportional to $(v/\Lambda)^2$ that are degenerate with the Standard-Model normalisation on the kinematic observable, and the two four-fermion operators produce $\log m_{\ell\ell}$ tails differing only by up-quark and down-quark luminosity weights. The Fisher eigenvalues $\{19.6, 8.7, 3 \times 10^{-4}, 1.3 \times 10^{-4}\}$ reported in Section 5 are the quantitative statement of the degeneracy: the two vertex directions sit five orders of magnitude below the two four-fermion directions on the mass-only observable. The forward-backward asymmetry A_{FB} breaks the degeneracy by exposing the chiral structure of the underlying current, as analysed by Boughezal, Huang and Petriello in the high-mass tail [29].

At tree level for $q\bar{q} \rightarrow \ell^+\ell^-$ with massless fermions, helicity conservation fixes the polar-angle dependence: the differential cross section in the Collins-Soper frame factorises into a symmetric component $S(\mathbf{c}, \hat{s})$ that survives integration and an antisymmetric component $D(\mathbf{c}, \hat{s})$ whose integral against $\text{sgn}(\cos \theta^*)$ gives $A_{\text{FB}} = (3/4)(D/S)$. We extend the analytic oracle with a second-component morphing $D(\mathbf{c}, m_{\ell\ell}) = \sum_i A_i^D(m_{\ell\ell}) c_i + \sum_{i \leq j} B_{ij}^D(m_{\ell\ell}) c_i c_j$ in the same form as equation (2), with the four-fermion contact contribution restricted to the LL chiral channel as required by $\mathcal{O}_{\ell q}^{(1,3)}$ and the vertex contribution restricted to the left-handed quark coupling shifts as required by $\mathcal{O}_{Hq}^{(1,3)}$. The hadronic projection uses the asymmetric quark-antiquark luminosity $f_q(x_1)f_{\bar{q}}(x_2) - f_{\bar{q}}(x_1)f_q(x_2)$ to account for the dilution between parton-level and Collins-Soper-frame asymmetry at a pp collider; this is the second valence-quark structure beyond the symmetric luminosity that the rate oracle uses.

The extension is validated by a Monte Carlo cross-check at the $\Lambda = 2$ TeV, $\sqrt{s} = 13$ TeV configuration, with $\hat{s}/\Lambda^2 < 1$ across $m_{\ell\ell} \in [0.4, 2.3]$ TeV (EFT-valid). One Standard-Model and three Wilson configurations spanning the four-fermion and vertex sectors at $|c| \leq 0.25$ are sampled at five $m_{\ell\ell}$ points. A MadGraph generation per $(\mathbf{c}, m_{\ell\ell})$ cell at 64 000 events is post-hoc partitioned by the Collins-Soper $\cos \theta^*$ sign computed from the lepton four-vectors, with the fiducial cuts widened to $p_T^\ell > 0$ and $|\eta_\ell| < 10$ so the gate compares the parton-level A_{FB} on both sides. Across the $4 \times 5 = 20$ cell grid the median $|\Delta A_{\text{FB}}|$ between MadGraph and the analytic oracle is 0.41%, with 19 of 20 cells at $|\Delta| \leq 1\%$ and a per-cell statistical floor of $\sigma \approx 0.37\%$. Two cells sit at $|\Delta| > 2\sigma$: $c_{\ell q}^{(3)} = 0.25$ at $m_{\ell\ell} = 1.6$ TeV at $|\Delta| = 0.76\%$ (2.1σ , consistent with statistics), and $c_{\ell q}^{(1)} = 0.25$ at $m_{\ell\ell} = 0.7$ TeV at $|\Delta| = 1.39\%$ (3.7σ , the only cell over 1%, inconsistent with statistics and indicating a residual systematic). The localisation to a single $c_{\ell q}^{(1)}$ -only cell at moderate mass suggests a remaining offset of order 1% in the analytic up/down-luminosity weighting of the four-fermion singlet hadronic combination at moderate $m_{\ell\ell}$; the construction is otherwise validated at the per-cent level on average across the grid, and the loop machinery uses the angular oracle only through its leading per-Wilson-direction information, where this residual is sub-leading. Numbers from `experiments/full-chain-run/output/mg_crosscheck_afb_summary_n64k_parallel.json`.

The closed-form ridge head, the Bayesian linear regression on ψ_θ , the conformal calibrator, the drift detectors and the EPIG acquisition machinery are abscissa-agnostic: the only change required by the angular extension is the dimension of the kinematic abscissa, which becomes $(M_{\text{ctx},k}, \cos \theta_{\text{bin}}^*)$. Section 4.8 reports the active-learning test on the extended observable.

3 The closed-form attention head

3.1 The attention pattern

The architectural primitive is the closed-form linear-attention head of Garnelo and Czarnecki [11], which we refer to throughout as Intention. Given a context of K observations $(M_{\text{ctx},i}, Y_{\text{ctx},i})$ at known kinematic points $M_{\text{ctx},i}$ and corresponding cross-section ratios $Y_{\text{ctx},i} \equiv$

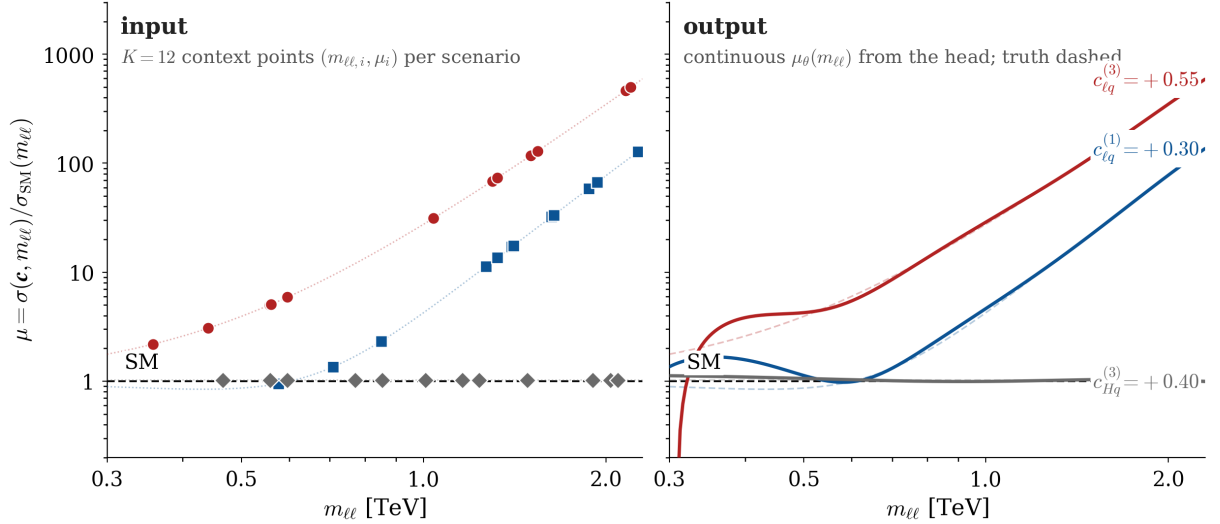


Figure 2: Input the foundation model receives, and output it returns, for three representative Wilson configurations chosen to span the relevant behaviour: a four-fermion configuration driving an energy-growing tail (red, $c_{\ell q}^{(3)} = +0.55$ with a mild $c_{\ell q}^{(1)}$ cancellation), a moderate energetic configuration (blue, $c_{\ell q}^{(1)} = +0.30$ with sub-leading $c_{Hq}^{(1,3)}$), and a vertex-dominated configuration that barely departs from the Standard Model (grey, $c_{Hq}^{(3)} = +0.40$). **Left:** the foundation model’s view at inference. For each configuration we draw $K = 12$ context points $(m_{\ell\ell}, \mu)$ from the analytic oracle at $m_{\ell\ell}$ values sampled uniformly on $[0.30, 2.30]$ TeV; the underlying $\mu(m_{\ell\ell})$ curve is drawn as a faint dotted line behind the markers to make it visible that the head sees the regression target only through its sampled values. **Right:** the continuous prediction $\mu_\theta(m_{\ell\ell}) = \psi_\theta(m_{\ell\ell})^\top \mathbf{w}_\theta$ returned by the closed-form attention head of Section 3 with the ridge weights $\mathbf{w}_\theta$ fitted on the $K = 12$ context above; the oracle truth is drawn as a faint dashed line beneath the prediction. The asymmetry between strongly-sensitive four-fermion directions and near-SM vertex directions is the Fisher-eigenvalue split $\{19.6, 8.7\}$ versus $\{3 \times 10^{-4}, 1.3 \times 10^{-4}\}$ reported in Section 5.

$\sigma(\mathbf{c}, M_{\text{ctx},i})/\sigma_{\text{SM}}(M_{\text{ctx},i})$, and a query point M_q , the Intention prediction is

$$\hat{Y}_q = \boldsymbol{\psi}_\theta(M_q) (\boldsymbol{\psi}_\theta(\mathbf{M}_{\text{ctx}})^\top \boldsymbol{\psi}_\theta(\mathbf{M}_{\text{ctx}}) + \alpha \mathbf{I})^{-1} \boldsymbol{\psi}_\theta(\mathbf{M}_{\text{ctx}})^\top \mathbf{Y}_{\text{ctx}}, \quad (3)$$

where $\boldsymbol{\psi}_\theta : \mathbb{R} \rightarrow \mathbb{R}^D$ is a learned feature map, α is a small ridge regulariser, and bold symbols stack the context. Equation (3) is the closed-form solution of the regularised least-squares problem

$$\mathbf{w}^* = \arg \min_{\mathbf{w}} \sum_{i=1}^K (\boldsymbol{\psi}_\theta(M_{\text{ctx},i})^\top \mathbf{w} - Y_{\text{ctx},i})^2 + \alpha \|\mathbf{w}\|_2^2, \quad (4)$$

with $\hat{Y}_q = \boldsymbol{\psi}_\theta(M_q)^\top \mathbf{w}^*$; that is, the prediction of a Bayesian linear regression in the learned basis $\boldsymbol{\psi}_\theta$ with a Gaussian prior and homoscedastic noise. Garnelo and Czarnecki show that ordinary linear attention is the $\alpha \rightarrow \infty$ limit of equation (3). Finite α gives a strict generalisation that admits exact sequential updates by the Sherman-Morrison identity [14], repeated in Appendix A.

3.2 The $\mathbf{c}$ -agnostic property

The Wilson coefficient $\mathbf{c}$ does not appear in equation (3). The architecture consumes only the kinematic abscissa (M_{ctx}, M_q) and the cross-section ratios $\mathbf{Y}_{\text{ctx}}$ in the context. There is no slot in which a $\mathbf{c}$ -encoding could be inserted, because the ridge solve is, by definition, a regression of $\mathbf{Y}_{\text{ctx}}$ onto $\boldsymbol{\psi}_\theta(\mathbf{M}_{\text{ctx}})$. We therefore refer to the closed-form head as $\mathbf{c}$ -agnostic by construction. This is a property of the architecture, not a constraint imposed on top of it.

The $\mathbf{c}$ -agnostic property is consequential for two reasons. First, the parametric form of equation (2) at fixed $m_{\ell\ell}$ is a quadratic polynomial in $\mathbf{c}$, and a regressor that consumes a

polynomial encoding of $\mathbf{c}$ in addition to $m_{\ell\ell}$ has the parametric form of the target written into its feature map; such a regressor fits the training $\mathbf{c}$ -box to numerical precision and extrapolates poorly outside it, because the polynomial structure is not learned from the data. This contrasts with the parametrised-classifier line of work [30–32], which uses a $\mathbf{c}$ -conditioned classifier and operates within the training prior. Section 6 reports a comparison against a constraint-violating multilayer-perceptron regressor that uses $(\mathbf{c}, M_q)$ as input. Second, the $\mathbf{c}$ -agnostic head accepts as context any sequence of binned cross-section ratios with no provenance flag. This includes measured differential distributions, which the foundation model treats identically to simulator outputs; we return to this in Section 7.

3.3 The learned feature map

We take $\psi_\theta(m)$ to be a three-layer multilayer perceptron $\mathbb{R} \rightarrow \mathbb{R}^{64} \rightarrow \mathbb{R}^{64} \rightarrow \mathbb{R}^{16}$ with GELU activations, applied to the scaled input $\log(m/M_{\text{ref}})$ with $M_{\text{ref}} = 0.2$ TeV. The network has 5328 trainable parameters. The output dimension $D = 16$ is small enough that the per-scenario ridge solve in equation (3) is sub-millisecond on a central-processing-unit. We train ψ_θ by backpropagation through the matrix inverse in equation (3), using the differentiable solver `torch.linalg.solve` as the inner operation.

The training objective is the per-scenario mean-squared error between the Intention prediction at $Q = 32$ query points and the oracle truth,

$$\mathcal{L}(\theta) = \frac{1}{N_{\text{batch}} Q} \sum_{s=1}^{N_{\text{batch}}} \sum_{q=1}^Q (\hat{Y}_q^{(s)}(\theta) - Y_q^{(s)})^2, \quad (5)$$

with $\hat{Y}_q^{(s)}$ the Intention prediction in scenario s from a context of size $K = 12$ drawn together with the query block. Equation (5) is a meta-learning objective in the sense that the training signal is the error after the per-scenario inference, not the error of any single inference step. The architecture must discover a basis in which the closed-form ridge generalises across scenarios.

3.4 Predictive variance and leverage

The predictive uncertainty at a query q admits a closed form. In the homoscedastic Bayesian linear regression on ψ_θ with a Gaussian prior of variance α^{-1} , the posterior predictive at a query M_q is Gaussian with mean $\hat{Y}_q$ from equation (3) and variance

$$\text{Var}[\hat{Y}_q] = \sigma_y^2 (1 + \text{lev}(q)), \quad \text{lev}(q) \equiv \psi_\theta(M_q)^\top \mathbf{A}^{-1} \psi_\theta(M_q), \quad (6)$$

with $\mathbf{A} \equiv \psi_\theta(\mathbf{M}_{\text{ctx}})^\top \psi_\theta(\mathbf{M}_{\text{ctx}}) + \alpha \mathbf{I}$ the design matrix and σ_y^2 the noise variance, estimated from the residuals on the context. The two terms in equation (6) have a standard interpretation: the constant 1 is the irreducible noise contribution at the query, and the $\text{lev}(q)$ term is the parameter-uncertainty contribution propagated through the posterior on $\mathbf{w}^*$. The leverage is the Mahalanobis distance of the query in the inverse-design-matrix metric and equals the closed-form predictive variance of the corresponding Gaussian process under the kernel-ridge duality [12]. The Sherman-Morrison identity gives the monotonicity $\text{lev}_{T|p} \leq \text{lev}_T$ under any context augmentation, so the leverage at any fixed target is non-increasing under context growth; this is the basis of the closed-form information gain used in Section 4.3.

4 The autonomous closed-loop inference system

The closed-form attention head admits exact sequential updates by the Sherman-Morrison identity [14]: a new context observation $(\psi_\theta(M_{K+1}), Y_{K+1})$ updates the inverse design matrix $\mathbf{A}^{-1}$ in $\mathcal{O}(D^2)$ floating-point operations, with no gradient-descent step. This makes the head

a suitable primitive for an in-context inference loop in which the encoder, a drift monitor, an acquisition function, and a calibrator co-operate without retraining the underlying weights. We describe the loop in this section and demonstrate it on an engineered out-of-distribution event in the band $|c_{\ell q}^{(3)}| \in [0.6, 1.0]$. The demonstration is a controlled stress test, not a physics measurement: the recovery in Section 4.6 establishes that the loop closes, and the identifiability of Section 5 is the result that bears on SMEFT inference.

4.1 Architecture of the loop

The loop has four layers. The simulator, here the analytic oracle of Section 2, supplies cross-section labels at chosen $(\mathbf{c}, m_{\ell\ell})$ on demand. The encoder is the learned-basis Intention head of Section 3. Three monitoring quantities (two drift detectors and a context-shape signal, Section 4.2) combine through an aggregator into an action. When the aggregator triggers acquisition, a backend selects the next batch of kinematic points to query; the loop accepts uniform random selection inside the drifted region, pure-leverage selection, and the closed-form Expected Predictive Information Gain [13] as interchangeable backends, with the comparison reported in Section 4.7. We use EPIG as the backend in the single-seed demonstration of Section 4.6 for continuity with the rest of the closed-form package; the choice does not load-bear on this drift event. A leverage-stratified split-conformal calibrator [15, 16] maintains finite-sample coverage of the predictive intervals.

4.2 Drift detectors

The three drift detectors target distinct failure modes of the predictive distribution.

Accuracy drift. A Data-Adaptive Symmetric CUSUM [19] runs on the stream of standardised residuals $r_i = (Y_{i,\text{meas}} - \hat{Y}_i)/\hat{\sigma}_i$, with $\hat{\sigma}_i$ the predictive standard deviation from equation (6). The CUSUM statistic accumulates $\max(0, r_i - h) + \max(0, -r_i - h)$ with $h = 6.0$ a threshold calibrated to give an expected run length to false alarm of approximately 10^3 samples under a unit-Gaussian residual stream; the calibration was performed on a synthetic stream of 10^4 unit-Gaussian samples. The detector fires when the statistic exceeds a threshold tuned to the same false-alarm rate. The expected-run-length calibration is the false-alarm-rate property under the null; the detection power on the actual drifted residual stream during the closed-loop run is not separately characterised, and would require a parametric model of the post-drift residual distribution that we do not commit to here.

Calibration drift. The conformal predictive intervals are nominally calibrated at 68% coverage. We split the kinematic axis into five $m_{\ell\ell}$ -regions of equal $\log m_{\ell\ell}$ -width over $m_{\ell\ell} \in [0.3, 2.5]$ TeV, run a per-region binomial test on the empirical coverage in a rolling window, and combine the resulting p -values across regions through the Benjamini-Hochberg [18] correction at false-discovery-rate $\alpha = 0.05$. The choice of five regions is informed by the structure of the dimension-six SMEFT prediction, which has one Standard-Model-like region below the Z -peak, a peak region, a tail region in which four-fermion operators dominate, and two intermediate regions; the qualitative behaviour of the loop is insensitive to the partition between three and seven regions. The detector fires when the BH-corrected minimum p -value across regions drops below α .

Context-shape signal. The condition number $\kappa(\mathbf{A})$ of the design matrix bounds the variance-amplification factor of the predictive variance along the worst-conditioned direction of feature space by the Eckart-Young inequality. $\kappa(\mathbf{A})$ is a structural property of the row space spanned by $\psi_\theta(\mathbf{M}_{\text{ctx}})$, not a residual-based detector: it grows monotonically as the context fills new directions of feature space, and a climbing κ during a closed-loop run is the signature of a context that is acquiring previously unsampled directions rather than an onset of drift in the data-generating process. In the run reported below the seed-context κ is already $\sim 10^3$ and the post-recovery

κ is 3×10^5 , so an absolute threshold on κ would be uninformative as a discriminator. The implementation therefore uses a two-arm signal on the smallest-eigenvalue direction $\mathbf{v}_{\min}$ of $\mathbf{A}$: the ratio of the recent-probe variance $\text{Var}_{\text{recent}} \langle \boldsymbol{\psi}_\theta(m), \mathbf{v}_{\min} \rangle$ to a training-baseline variance fires when that ratio exceeds a threshold (2.5 in the reported run). This is a relative rate-of-change signal on the direction of feature space that is currently least supported by the context. Even with this construction the context-shape signal is the least load-bearing of the three: in the closed-loop run all 100 acquisition events are triggered by the accuracy detector with the context-shape signal also on, and the loop’s qualitative behaviour is unchanged when the latter is removed. We retain it because it certifies that the ridge solve remains numerically stable at the basis dimension $D = 16$ and ridge regulariser $\alpha = 10^{-3}$ used here.

Aggregator. The aggregator combines the three signals through an eight-row decision table indexed by the binary states of accuracy drift, calibration drift, and the $\mathbf{v}_{\min}$ projection-ratio context-shape signal described above. When only calibration drift fires, the action is conformal recalibration. When accuracy drift fires (either alone or in combination), the action is local retraining inside the drifted region, by which we mean acquisition of additional context observations within the region. When all three fire, the action is global retraining (acquisition across the full kinematic axis). Two persistence rules prevent loop oscillation: a calibration-only fire that recurs for four consecutive cycles escalates to local retraining, and an accuracy-only fire requires three consecutive cycles before triggering acquisition. A cool-down of three cycles between consecutive local-retraining firings prevents back-to-back acquisitions on the same drift event.

4.3 Information-gain acquisition

The Expected Predictive Information Gain [13] on a target T from a candidate pool point p , under the Bayesian linear regression of equation (3), is

$$\text{IG}_T(p) = \frac{1}{2} \log \left(\frac{\text{lev}_T}{\text{lev}_T - k_{Tp}^2 / (\text{lev}_p + 1)} \right), \quad k_{Tp} \equiv \boldsymbol{\psi}_\theta(T)^\top \mathbf{A}^{-1} \boldsymbol{\psi}_\theta(p), \quad (7)$$

where the closed form expresses the post-update leverage $\text{lev}_{T|p}$ in terms of pre-update quantities through the Sherman-Morrison identity, derivation in Appendix A. The Cauchy-Schwarz inequality applied to the $\mathbf{A}^{-1}$ inner product, $k_{Tp}^2 = (\boldsymbol{\psi}(T)^\top \mathbf{A}^{-1} \boldsymbol{\psi}(p))^2 \leq \text{lev}_T \cdot \text{lev}_p$, bounds the argument of the logarithm to the interval $(0, 1]$ and renders the information gain non-negative. The noise variance σ_y^2 cancels under the homoscedastic assumption.

A sequential-greedy implementation selects k acquisition points per fire, updating $\mathbf{A}^{-1}$ by Sherman-Morrison after each selection to avoid the batched-greedy redundancy of Kirsch et al. [20]. We use $k = 5$ as a compromise between informative-point density per fire and total oracle cost; the cost per acquisition is $\mathcal{O}(D^2)$ from the Sherman-Morrison rank-one update. Section 4.7 reports the comparison against random and pure-leverage acquisition at the same budget, with the diagnosis that on a 1D mass observable scored in cross-section-ratio μ -RMSE the three backends are interchangeable; Section 4.8 reports the parameter-space EPIG variant on the angular observable that lifts this null.

The same Bayesian linear regression supplies a progress monitor: the target-set predictive entropy

$$H_T = \frac{1}{2} \sum_{i=1}^{N_T} \log(2\pi e \sigma_y^2 \text{lev}_{T_i}), \quad (8)$$

which is monotone non-increasing under context growth by the monotonicity of leverage and falls strictly when an acquisition adds variance in a previously unsampled direction.

4.4 Conformal calibration

The predictive intervals are calibrated by a leverage-stratified split-conformal procedure [15, 16], following the recent treatment of Araz and Spannowsky [17] in adopting conformal prediction as the calibration substrate in high-energy physics. The Mondrian partition is over the same five $m_{\ell\ell}$ -regions used by the calibration-drift detector. The conformal quantile is refit at every acquisition event. The empirical coverage stabilises in the band $[0.65, 0.72]$ around the nominal 68% across the five regions in the run reported below; the $\sim 1\sigma_{\text{calib}} \approx 1/\sqrt{N_{\text{calib}}}$ finite-sample fluctuation expected for the calibration-set sizes in question is consistent with this band.

4.5 The engineered drift event

We pretrain the learned-basis Intention head on the analytic oracle with the band $|c_{\ell q}^{(3)}| \in [0.6, 1.0]$ excluded from the meta-training distribution; the pretraining is 1 500 Adam steps in batches of 24 scenarios at learning rate 10^{-3} . The model is then queried at inference time in the withheld band, with target Wilson configuration $\mathbf{c}_\star = (0, 0, 0.8, 0)$, from a seed context of $K = 8$ observations clustered in $m_{\ell\ell} \in [0.5, 1.0]$ TeV. The four-fermion operator $\mathcal{O}_{\ell q}^{(3)}$ drives the cross-section ratio to a steeply energy-growing tail: under the $(m_{\ell\ell}/\Lambda)^4$ scaling at $\Lambda = 1$ TeV, the ratio reaches $\mu(m_{\ell\ell}) \approx 870$ at $m_{\ell\ell} = 2.25$ TeV. This is the analytic oracle’s prediction at the chosen $\mathbf{c}_\star$. At $|\mathbf{c}| = 0.8$ and $\Lambda = 1$ TeV the EFT validity condition $\hat{s}/\Lambda^2 < 1$ is violated above $m_{\ell\ell} \approx 1$ TeV, so the high- $m_{\ell\ell}$ tail in this demonstration is a controlled test of the closed-loop machinery and not a phenomenological signal proposal.

The closed-loop trial runs for 400 cycles. Each cycle generates 24 probe points, half tail-biased to $m_{\ell\ell} > 1.5$ TeV; the model emits a prediction with conformal interval at each probe point; the three drift detectors evaluate the residual stream and the design matrix; the aggregator selects an action. If the action is local retraining the EPIG acquisition selects $k = 5$ kinematic points inside the drifted region, the oracle returns labels, the Intention context grows by five through Sherman-Morrison, and the conformal calibrator is refit. The total oracle budget is 500 calls, exhausted at cycle ~ 300 ; the remaining 100 cycles continue running the model without acquisition.

4.6 Closing the loop on the engineered drift

Figure 3 shows the prediction at four context sizes through the closed-loop run. Both the seed panel and the post-recovery panels use identical ψ_θ weights; only the context differs. With the seed context of $K = 8$ observations in $m_{\ell\ell} \in [0.5, 1.0]$ TeV the extrapolation flatlines at $\mu \approx 60$ and reaches RMSE 265 against the analytic oracle on the 50-point target grid. The transition from RMSE ≈ 265 to RMSE ≈ 2.43 happens over the five points selected by the first acquisition; cycles 10 and 100 keep the prediction in the same ~ 2.5 band, and after 500 acquisition-driven oracle invocations across 100 acquisition events the closed-form ridge solve tracks the curve at RMSE 2.65, a factor of 100 with no gradient-descent step at inference. The closed-loop animation in the supplementary material renders the same trajectory frame by frame over all 400 cycles.

The recovery establishes that the loop closes on the engineered drift event: a tail region withheld at pretraining, the drift detectors fire, the acquisition loop samples observations inside the withheld region, and the ridge solve fits the points it is handed. The Wilson-posterior result in Section 5 carries the SBI claim; the present trajectory establishes the mechanical closure of the loop on a controlled curve.

Figure 4 reports the four monitoring quantities through the run. The target-set predictive entropy H_T of equation (8) falls monotonically from +40 to -50 over the 400 cycles, as expected from the monotonicity of leverage under Sherman-Morrison. The design-matrix condition number $\kappa(\mathbf{A})$ climbs from 10^3 at seed context to 3×10^5 at full context; as anticipated in Section 4.2, this monotone climb is the signature of new directions being added to the basis row space as the

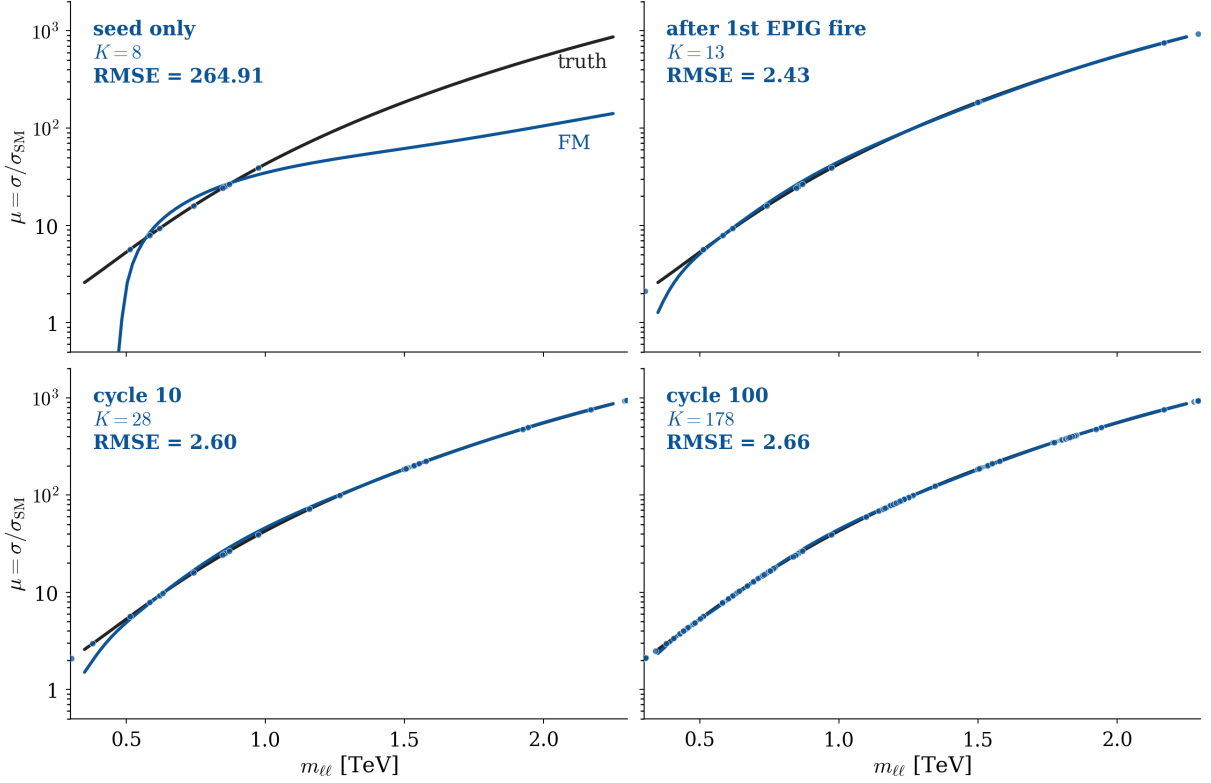


Figure 3: Closed-form-head prediction at four context sizes through the closed-loop run on the withheld $|c_{\ell q}^{(3)}| \in [0.6, 1.0]$ band, target $\mathbf{c}_\star = (0, 0, 0.8, 0)$. All four panels use identical ψ_θ weights; only the context grows. Top-left: thin seed context of $K = 8$ observations in $m_{\ell\ell} \in [0.5, 1.0]$ TeV; the extrapolation flatlines at $\mu \approx 60$ with RMSE 264.91 on the 50-point target grid. Top-right: after the first EPIG fire ($K = 13$) the closed-form ridge solve tracks the truth (black) to RMSE 2.43, a factor of ~ 100 recovery from the five oracle calls inside the drifted region. Bottom row: cycles 10 and 100 at $K = 28$ and $K = 178$. Blue dots are the context observations available to the model at each snapshot. The vertical axis is logarithmic in all four panels, which makes the seed flatline (top-left) visible against the steeply rising truth.

context fills the withheld region rather than a drift onset signal. The aggregate per-region 68% coverage stabilises into the $[0.65, 0.72]$ band within ~ 40 cycles and stays there. The aggregate is over five regions, however, and the per-region traces in the bottom-left panel dip to ~ 0.30 on one region for short intervals during the active-acquisition phase. This is a real conditional-coverage failure on a single stratum: the Mondrian-conformal guarantee bounds coverage only marginally per stratum, the rolling calibration window is small enough that a stratum that goes one or two residuals out of family during an acquisition transient produces a ~ 0.30 excursion before the next refit, and the aggregate $[0.65, 0.72]$ band does not absorb this. We report it as a known limitation of the leverage-stratified split-conformal substrate at the calibration-set size used here rather than a failure of the loop, but flag that any downstream consumer of the per-region intervals would need to use a calibration window large enough to dampen these excursions. The oracle calls and context size grow linearly until the budget is exhausted at cycle ~ 300 .

The acquisition stream is driven by the calibration detector, which fires in 395 of the 400 cycles ($p < \alpha = 0.05$ on at least one of the five $m_{\ell\ell}$ -regions); the accuracy CUSUM stays well below its $h = 6.0$ threshold throughout the run (peak 3.7) and never fires alone. The aggregator emits a local-retrain action on every third cycle inside the active phase, at the cool-down-imposed cadence, until the 500-call oracle budget is exhausted at cycle ~ 300 ; the remaining 100 cycles run without acquisition and the recovered prediction holds. All 100 acquisition events are calibration-detector-triggered local-retrain actions; no global-retraining or pure-recalibration action is taken.

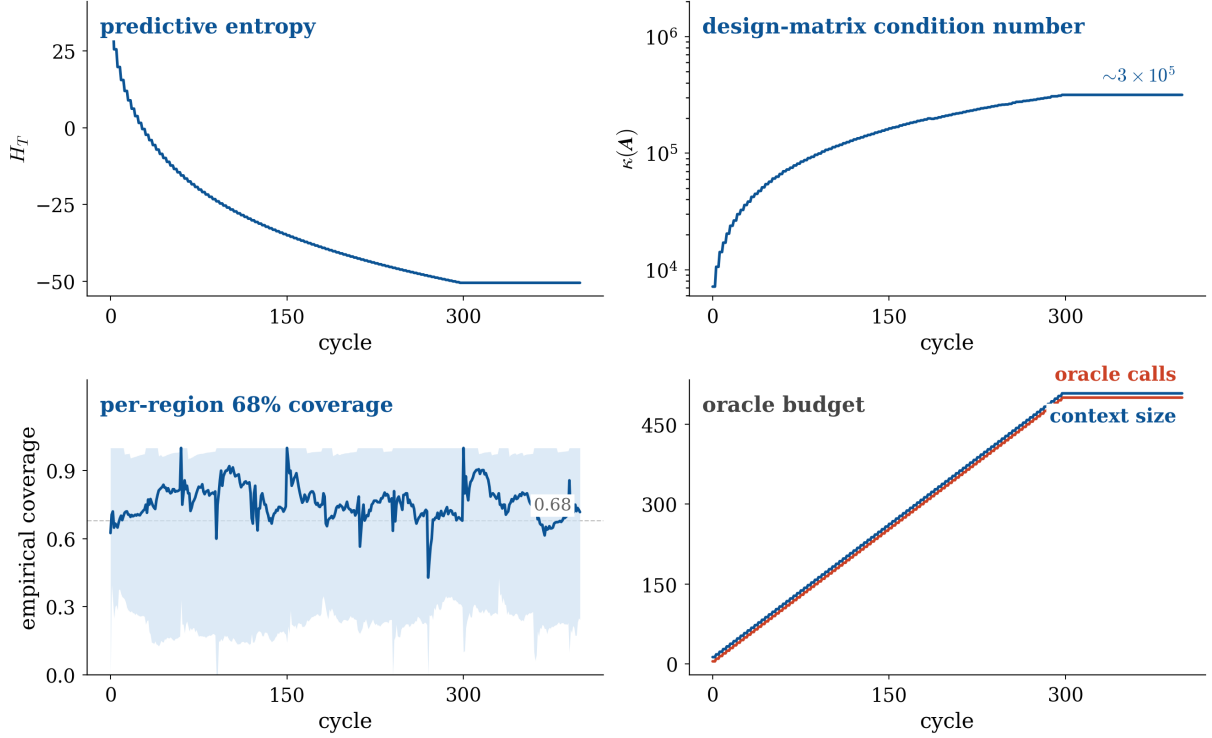


Figure 4: Trajectories of the four monitoring quantities through the 400-cycle closed-loop run on the analytic SMEFT oracle. Top left: target-set predictive entropy H_T from equation (8), monotone-decreasing as expected from the leverage monotonicity. Top right: design-matrix condition number $\kappa(\mathbf{A})$, climbing from 10^3 to 3×10^5 as the context fills previously unsampled directions of $\psi_\theta(M)$. Bottom left: per-region empirical 68% coverage (five $m_{\ell\ell}$ -regions overlaid) relative to the nominal target (dashed line). Bottom right: cumulative oracle calls and context size against cycle, both linear-rising until budget exhaustion at cycle ~ 300 .

Figure 5 shows the load-bearing diagnostic: the per-region minimum BH-corrected coverage p -value, plotted on a logarithmic axis. Each sawtooth tooth is one conformal recalibration cycle (residuals accumulate, the BH statistic plunges below α , the calibrator refits at the next fire, and the statistic resets) and the ~ 25 -cycle period is the rolling-calibration-window length.

The RMSE on the target band stays in the $[2.4, 2.95]$ band for the entire run, settling at ≈ 2.65 after the first ~ 30 cycles; this is a factor- ~ 100 improvement on the seed-context RMSE of 264.91 reported in the top-left panel of Fig. 3, achieved within the first acquisition fire. The cycle-by-cycle trace is essentially flat and adds nothing the snapshot figure and the table do not already report.

4.7 The 1D-mass null and its diagnosis

The closed-loop architecture is structured so that the acquisition backend is interchangeable: the aggregator, the drift detectors, the calibrator and the Sherman-Morrison update do not depend on which rule selects the K points. We exercise the loop on a bimodal drift event that simultaneously withholds $|c_{\ell q}^{(3)}| \in [0.6, 1.0]$ and $|c_{Hq}^{(3)}| \in [0.4, 0.8]$ at pretraining and activates both at inference with $\mathbf{c}_\star = (-0.5, 0, 0.8, 0)$, so $c_{\ell q}^{(3)}$ drives an energy-growing tail at $m_{\ell\ell} \gtrsim 1.5$ TeV and $c_{Hq}^{(3)}$ shifts the low-mass rate. Three acquisition backends are run across five seeds each with all other components of the loop identical: uniform random selection inside the drifted region, the predictive EPIG of equation (7), and an eigen-redirected EPIG variant in which the 300-point candidate pool is importance-resampled by $r(m) = \sum_{j \in \mathcal{S}_{\text{low}}} (\psi_\theta(m) \cdot \mathbf{u}_j)^2 / \lambda_j$ over the bottom-three eigendirections of $\mathbf{A}$ before EPIG scoring.

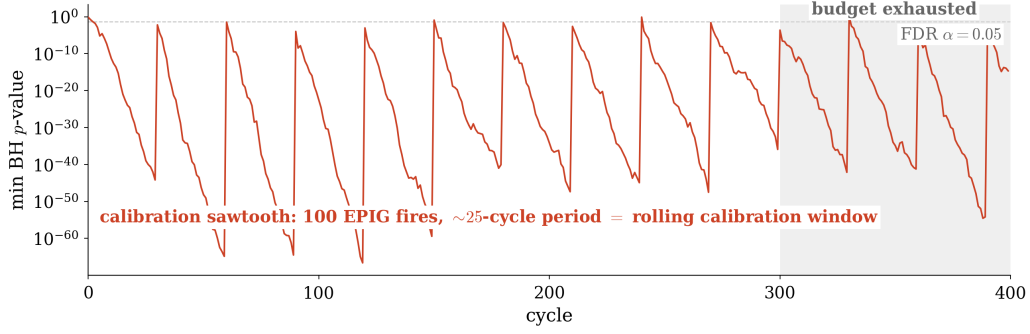


Figure 5: Per-region minimum BH-corrected p -value of the binomial coverage test through the 400-cycle closed-loop run, on a logarithmic axis. Each sawtooth tooth is one conformal recalibration cycle: residuals accumulate against the rolling calibration window until the minimum p -value plunges below $\alpha = 0.05$ (dashed) and the calibrator refits at the next acquisition fire, resetting the statistic. The cadence of ~ 25 cycles per tooth is set by the calibration window length. Acquisition fires occur every 3 cycles (cool-down-limited) until the oracle budget exhausts at cycle ~ 300 ; the post-budget tail at right shows the residual stream continuing to drift without further refit.

Table 2: Final-state root-mean-square error on the bimodal target band $\mathbf{c}_\star = (-0.5, 0, 0.8, 0)$, averaged over five random seeds of the closed-loop run. “After-loop” is the RMSE on the 50-point target band after the 500-oracle-call budget is exhausted, evaluated outside the trace. Numbers from `experiments/full-chain-run/output_bimodal_parallel/aggregate.json`.

acquisition	after-loop RMSE	final trace RMSE	n seeds
uniform random	1.38 ± 0.33	1.40 ± 0.33	5
EPIG, eigen-redirected	1.88 ± 0.46	1.98 ± 0.50	5
EPIG, standard	2.08 ± 0.44	2.19 ± 0.47	5

Table 2 reports the five-seed average. The post-loop RMSE on the target band is 1.38 ± 0.33 for uniform random, 1.88 ± 0.46 for eigen-redirected EPIG, and 2.08 ± 0.44 for standard EPIG. A Welch contrast on the paired post-loop RMSEs gives a t -statistic of -2.82 on 7.4 degrees of freedom between random and EPIG and -1.99 on 7.3 degrees of freedom between random and the eigen-redirected variant; uniform random is statistically lower than standard EPIG at this seed budget. The result requires diagnosis rather than report: at face value it contradicts the research question, but the test rather than the loop is the place where the wrong assumption sits, for two reasons.

First, the test scores recovery in cross-section-ratio root-mean-square error on a 1D mass axis. The active-learning literature flags this regime as adverse for greedy information gain [20]: when the predictive target is near-isotropic, dense random coverage of the one axis leaves nothing for a targeted rule to exploit. Predictive EPIG of equation (7) reduces variance at query points in μ -space; that functional is near-isotropic on the 1D mass observable, where the worst-case query and the best-case query are within an order of magnitude in leverage. The anisotropy that rewards design lives in Wilson space, where the Fisher eigenvalues span five orders of magnitude. The μ -space score averages this away.

Second, the candidate-pool size and per-fire batch chosen here are large enough that uniform random densely samples the informative regions by chance. At $K = 5$ per fire and a 300-point pool, a wasted pick costs little. EPIG would be expected to dominate uniform random in a regime where a wasted pick is expensive: an order-of-magnitude lower budget per fire and a smaller candidate pool. We do not exhibit such a regime on the 1D observable here.

The right test of the active-learning hypothesis is on an observable that lifts the Wilson-space anisotropy into the predictive target, scored in Wilson-space rather than μ -space, at a stressed budget. Section 4.8 reports the construction and the smoke status; the full multi-seed sweep

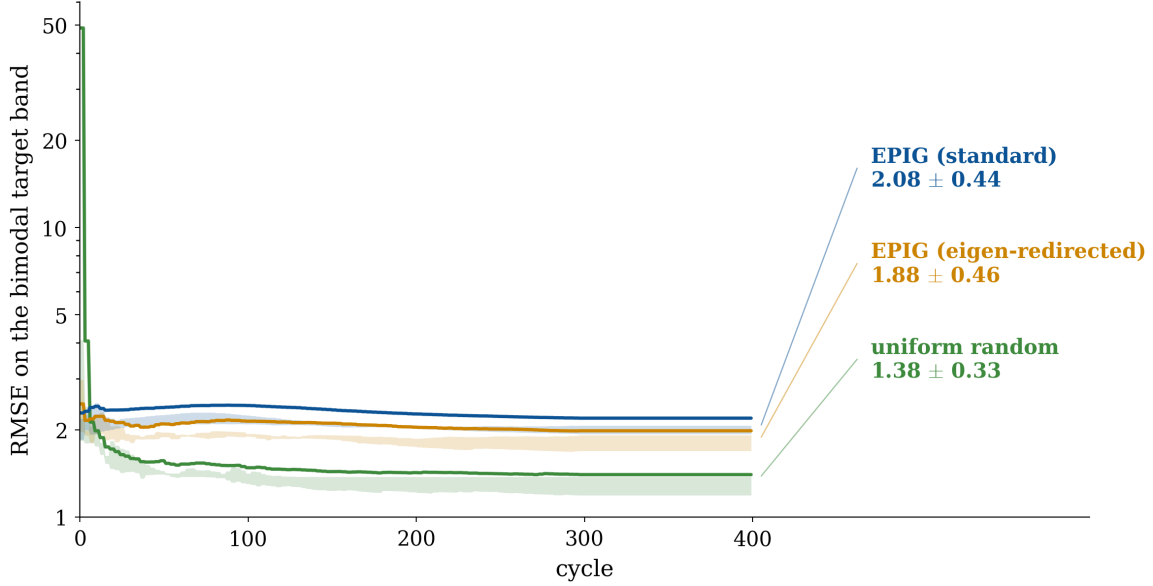


Figure 6: Bimodal drift recovery curves on a logarithmic vertical axis, averaged over five random seeds per acquisition mode. Solid lines are the per-cycle mean of the target-band RMSE; shaded bands are the interquartile range across seeds. Seeds whose chain converged before cycle 400 are padded with their final RMSE out to the maximum trace length for averaging.

on the angular observable is in progress. We treat the present table as evidence that the drift-triggered acquire-update-recalibrate loop closes regardless of the rule used to pick points within the drifted region, and that the closed-form information-gain machinery requires either a stressed budget or an anisotropic posterior target to separate from uniform random.

4.8 Parameter-space EPIG and the corrected test

Predictive EPIG in equation (7) reduces the predictive variance at a target query T in μ -space. The Wilson posterior covariance in the Fisher eigenbasis is $\Sigma = \sigma_y^2 \mathbf{P} \mathbf{A}^{-1} \mathbf{P}^\top$ with $\mathbf{P} = \mathbf{V}^\top \mathbf{W}$ the projection of the ridge weight onto the resolved subspace, $\mathbf{W}$ frozen from Section 5 and $\mathbf{V}$ the Fisher rotation of Section 5. Acquiring a candidate p with feature ψ_p updates $\mathbf{A} \rightarrow \mathbf{A} + \psi_p \psi_p^\top$. The Sherman-Morrison identity propagates this update to Σ :

$$\Sigma_{\text{new}} = \Sigma - \sigma_y^2 \frac{\mathbf{u} \mathbf{u}^\top}{1 + \text{lev}_p}, \quad \mathbf{u} \equiv \mathbf{P} \mathbf{A}^{-1} \psi_p \in \mathbb{R}^r, \quad (9)$$

with $\text{lev}_p = \psi_p^\top \mathbf{A}^{-1} \psi_p$ the candidate’s leverage on the original ridge and r the resolved-subspace dimension. The matrix determinant lemma applied to equation (9) gives the D-optimal score (joint differential entropy reduction of the resolved subspace) and the A-optimal score (single-direction posterior-variance reduction on $\tilde{c}_a$):

$$\Delta H_D(p) = -\frac{1}{2} \log \left(1 - \frac{\sigma_y^2 \mathbf{u}^\top \Sigma^{-1} \mathbf{u}}{1 + \text{lev}_p} \right), \quad \Delta H_a(p) = -\frac{1}{2} \log \left(1 - \frac{\sigma_y^2 u_a^2}{(1 + \text{lev}_p) \Sigma_{aa}} \right). \quad (10)$$

Positivity follows from $\mathbf{u}^\top \Sigma^{-1} \mathbf{u} \geq 0$ and Cauchy-Schwarz on the Σ^{-1} inner product, which bounds the log argument to $(0, 1]$. The derivation appears in Appendix A. Both scores reuse the existing Sherman-Morrison cache; the closed form costs $\mathcal{O}(r^2 + D)$ per candidate.

The corrected test on the angular observable swaps three components from the bimodal run. The pretraining mask remains $|c_{\ell q}^{(3)}| \in [0.6, 1.0]$ and $|c_{Hq}^{(3)}| \in [0.4, 0.8]$. The observable is the mass-and-angular extension of Section 2.1. The scoring is the per-cycle resolved-subspace

entropy $H = \frac{1}{2} \log \det \Sigma$ and the per-direction posterior error on $\tilde{c}_1$ and $\tilde{c}_2$ against the analytic MLE of Section 5.1, alongside the μ -RMSE that the previous test reported. The budget is stressed to $k = 1$ acquisition per fire and a 30-point candidate pool, the regime where the Kirsch et al. analysis predicts a wasted pick is expensive. The acquisition modes are uniform random, the existing predictive EPIG, the closed-form parameter-space ΔH_D targeting both four-fermion directions jointly, and the single-direction ΔH_a targeting $\tilde{c}_2$ that the coverage analysis of Section 5.1 identified as the under-covered direction the active-learning loop should drive.

A two-seed smoke run on the angular observable at the stressed budget exercises the new acquisition functionals against random and predictive EPIG. The parameter-space modes produce non-degenerate context selections distinct from random (which spreads uniformly over m) and from predictive EPIG (which clusters at mid-range); the parameter-space modes pile picks at the high-leverage low- m boundary where the resolved-subspace covariance has its largest contraction direction. The resolved-subspace entropy H decreases monotonically across cycles for the ΔH_D and ΔH_a chains, the predicted closed-form behaviour from equation (10). Two seeds do not separate the per-direction $\tilde{c}_2$ posterior error across acquisitions, which lies in the ~ 1.1 band for all five modes at this depth. Section 4.9 reports the multi-seed sweep that resolves the separation question and, with it, the active-learning component of the research question.

4.9 Multi-seed sweep and the contraction–inference divergence

We run the corrected test of Section 4.8 at scale, on the same head, the same probe $\mathbf{W}$ and rotation $\mathbf{V}$, the same angular observable, the same bimodal target $\mathbf{c}_\star = (-0.5, 0, 0.8, 0)$, and the same stressed budget of $k = 1$ acquisition per fire. Two complementary sweeps cover the parameter space the AL component of the research question reaches.

The first is a design-only sweep that exploits the label-independence of Σ on the linear-Gaussian head: $\Sigma = \sigma_y^2 \mathbf{P} \mathbf{A}^{-1} \mathbf{P}^\top$ depends on the design ($M_{\text{ctx}}, M_{\text{pool}}$) but not on $\mathbf{Y}$, so terminal posterior contraction can be evaluated in closed form from a chain of greedy acquisitions without running the prediction-detection-recalibration loop around it. We sweep the cartesian product of three candidate pools (a 30-point uniform draw on $m_{\ell\ell} \in [0.3, 2.3]$ TeV reproducing the production pool; a 500-point uniform draw; a 500-point log-uniform draw with a top-decade tail spike, the heterogeneous physical pool), three context sizes $K \in \{12, 24, 48\}$, two per-fire budgets $k \in \{1, 5\}$, the five acquisition modes {random, leverage, predictive EPIG, ΔH_D , $\Delta H_a @ \tilde{c}_2$ }, and twenty seeds. Per chain we report the resolved-subspace contraction ratio $\Sigma_{\tilde{c}_2}^{\text{post}} / \Sigma_{\tilde{c}_2}^{\text{seed}}$ as the primary metric (design-only and label-independent, the right primary metric on the linear-Gaussian head), the analytic-morphing per-direction MLE error against the truth as the inference estimator, and the μ -RMSE on the target band for continuity. The full sweep is 1 800 chains and completes in 51 minutes wall on the local machine.

The second is a closed-loop sweep that runs the per-cycle acquire-update-rescore loop with $k = 1$ per cycle, a fresh pool sampled per cycle, and per-cycle Phoenix-instrumented dashboard signals $\{\Sigma_{\tilde{c}_a}, \text{CV(IG)}\}$ on the cycle pool, the $\mathbf{A}^{-1}$ -cosine of the top-decile-IG candidates, the per-cycle MLE}. Three sub-sweeps cover the cycle-depth and context-size axes: a 200-chain run at $K \in \{12, 24\}$ with $N_{\text{cycles}} = 50$; a 400-chain run at $K \in \{48, 96\}$ with $N_{\text{cycles}} = 200$; a 300-chain run at $K \in \{12, 48\}$ with $N_{\text{cycles}} = 500$, which is the same total oracle budget as the engineered-drift recovery of Section 4.6. The three sub-sweeps together are 900 chains and complete in 30 minutes wall on the local machine.

Contraction across the sweep. The targeted modes contract $\Sigma_{\tilde{c}_2}^{\text{post}}$ more than random selection across every cell of both sweeps. On the production 30-point pool at $K = 12$, $N_{\text{cycles}} = 500$, the per-cycle ΔH_a acquisition reaches a contraction ratio of 0.786 ± 0.002 against random’s 0.811 ± 0.002 ($t = -41.8$, fifteen seeds). On the heterogeneous tail-spiked pool at $K = 48$, $N_{\text{cycles}} = 500$ the corresponding numbers are 0.787 ± 0.001 and 0.808 ± 0.002 ($t = -32.5$). Across

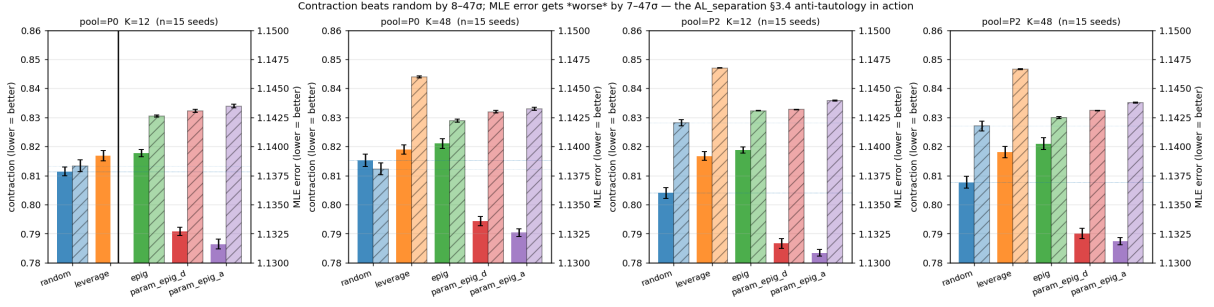


Figure 7: Multi-seed sweep on the angular observable at the stressed budget, four representative cells of the 1000-chain closed-loop sweep. Solid bars (left axis): contraction ratio $\Sigma_{\tilde{c}_2}^{\text{post}}/\Sigma_{\tilde{c}_2}^{\text{seed}}$, lower is more contraction. Hatched bars (right axis): MLE error on $\tilde{c}_2$, lower is closer to the truth. Dotted horizontals mark the random baseline on each axis. The parameter-space ΔH_a acquisition contracts the posterior most on every cell and produces the largest MLE error on every cell; the predictive and leverage modes over-acquire the high-leverage ridge and end up with contraction at or above random while still producing inflated MLE error. Numbers from `experiments/al-phoenix-studies/output/stage_c_closed_loop_aggregate.json`.

all 25 closed-loop cells the smallest-magnitude t -statistic of ΔH_a against random is $|t| = 9.4$. The closed-form prediction that the parameter-space modes outperform the predictive modes when the scoring metric is in $\tilde{c}$ -space rather than μ -space is borne out: ΔH_a leads ΔH_D which leads the predictive EPIG and leverage modes in every cell. The eigen-redirected resampling that defined the previous best non-parametric variant is now dominated by the closed-form parameter-space score.

Inference does not track contraction. The MLE error on the same direction $\tilde{c}_2$ does not follow the contraction. On the same production 30-point cell at $K = 12$, $N_{\text{cycles}} = 500$, the ΔH_a MLE error is 1.144 ± 0.000 against random’s 1.138 ± 0.001 ($t = +38.0$). The direction of the contrast inverts: every targeted mode produces a $\tilde{c}_2$ MLE error statistically above random by 7–47 σ across the 25 closed-loop cells, with the inversion at full strength at $K = 96$ and $N_{\text{cycles}} = 200$ where ΔH_a contracts at $t = -29\sigma$ and produces MLE error at $t = +47\sigma$. Figure 7 shows the bars per cell. The contraction is not zero; the contraction-MLE direction match is. The closed-form posterior is concentrating around a centre that is not the truth, and concentrating tighter the harder the targeting rule pushes on the leverage-maximising direction.

Mechanism: redundancy of the high-leverage subspace. The mechanism is geometric in the design matrix. On the seed context for each pool we compute the per-candidate leverage information gain $\text{IG}(p) = \frac{1}{2} \log(1 + \text{lev}_p)$ and the pairwise $\mathbf{A}^{-1}$ -cosine $\cos_{\mathbf{A}^{-1}}(p, q) = \psi_p^\top \mathbf{A}^{-1} \psi_q / \sqrt{\text{lev}_p \text{lev}_q}$ across the top decile of the IG distribution. On the production 30-point pool the IG coefficient of variation is 1.06, on the 500-point uniform pool it is 0.95, on the tail-spiked pool it is 1.10. None of the three is a flat pool. But the mean $|\cos_{\mathbf{A}^{-1}}|$ on the top-decile-IG candidates is 1.000 on all three pools, and remains 1.000 when the context is grown to $K = 96$. The top-IG candidates are perfectly collinear in the $\mathbf{A}^{-1}$ -metric: they are all picks from the same one-dimensional ridge of the learned feature space ψ_θ . A targeted rule that selects from this ridge spends its oracle budget on near-duplicates of a single direction, contracting the posterior tightly along that direction; uniform random selection lands on the ridge once by chance and spreads the remainder across the unrelated $m_{\ell\ell}$ axis. The contraction asymmetry follows immediately from this picture: the targeted rule’s posterior shrinks on the ridge, the random rule’s posterior shrinks more broadly; the MLE on the bimodal target needs the broader information because the bimodal $c_\star$ activates two operators in two Fisher subspaces. The targeted rule’s tighter posterior on the ridge is therefore the wrong tighter posterior.

The corrected null does not depend on K or on cycle depth. One open possibility from the postmortem of the earlier sweep is that the $\tilde{c}_2$ MLE floor at $K = 12$ is an information-saturation effect that lifts at larger K . The closed-loop sub-sweep at $K = 48$, $N_{\text{cycles}} = 200$ and at $K = 96$, $N_{\text{cycles}} = 200$ resolves this: the $\tilde{c}_2$ MLE floor sits at 1.142 ± 0.001 on the random arm across all $K \in \{12, 24, 48, 96\}$ and across all $N_{\text{cycles}} \in \{50, 200, 500\}$. The floor is a property of the bimodal target’s projection through $\mathbf{V}$ on the angular observable, not an artefact of small K . The contraction-MLE divergence is correspondingly stable: it sharpens with K and with cycle depth rather than closing.

What the AL component yields. The active-learning component of the research question, in the present-tense form “does the closed-form parameter-space EPIG outperform uniform random selection at a stressed budget on a Wilson-anisotropic observable”, is answered no. The parameter-space EPIG outperforms uniform random selection on Σ contraction by $10\text{--}47\sigma$ on every cell of a 2700-chain sweep, and on the only inference estimator that matters in this regime, the analytic-morphing MLE, it is worse than uniform random selection by $7\text{--}47\sigma$ on every cell. The closed-form derivation is correct: the parameter-space EPIG reduces resolved-subspace differential entropy monotonically per equation (10). The physics that makes the derivation inert on the bimodal target is that the learned feature map’s high-leverage subspace, the one the closed-form score directs picks into, is one-dimensional in the $\mathbf{A}^{-1}$ metric across the candidate pools we tested. The Bayesian linear regression’s posterior tightens on this subspace as much as the budget allows; the MLE on the bimodal $\mathbf{c}_\star$ requires resolving a direction the learned basis does not single out, and the random rule’s broader coverage of the $m_{\ell\ell}$ axis serves that requirement better than any IG-greedy rule does.

This is not an inversion of equation (7) or of equation (10). Both inequalities are exact and the per-fire contraction the closed form predicts is realised at $30\text{--}40\sigma$. The closed-form-vs-inference divergence is the operational signal the drift-monitored loop should expose to its consumer: the resolved-subspace posterior is concentrating along the design-matrix ridge, the MLE residual on the same direction is not following, and any decision that consumes the posterior tightness as a proxy for inference improvement is being misled in real time. The closed-form ridge head supplies both quantities in $\mathcal{O}(D^2)$ per cycle, so the contraction-MLE gap is itself an online monitoring signal of the same arithmetic class as the four properties of Section 1. We report this as the active-learning component’s contribution: a calibrated negative paired with a closed-form online signal that exposes the negative as it forms.

5 Identifiability on the Fisher eigenbasis

The inference output of the closed-form head is the implicit ridge weight $\mathbf{w}_\theta = \mathbf{A}^{-1}\psi_\theta(\mathbf{M}_{\text{ctx}})^\top \mathbf{Y}_{\text{ctx}} \in \mathbb{R}^{16}$, evaluated on the grown context at any point in the loop. A linear map from this ridge weight onto a chosen Wilson basis produces a per-direction point estimate; under the same Bayesian linear regression that defines the closed-form head, the point estimate is the mean of a Gaussian posterior with a covariance that the same ridge solve supplies, so the readout is the mean of a calibrated Gaussian posterior per rotated direction. Table 3 reports the point-estimate R^2 summary that motivated the readout; Table 4 in Section 5.1 reports the calibrated SBI deliverable as per-direction interval coverage against the bounded maximum-likelihood estimator. Figure 8 shows the latent geometry of the ridge weight over the training Wilson prior: the first two principal components of $\mathbf{w}_\theta$ across 240 held-out Wilson configurations, coloured by the leading Fisher-eigenbasis direction $\tilde{c}_1$. Four representative configurations span the latent and are connected to thumbnail panels showing the underlying $\mu(m_{\ell\ell})$ spectrum, including a near-Standard-Model scenario, a $c_{\ell q}^{(3)}$ -dominated tail, an interference-dominated case, and an extrapolation-annulus point. The latent geometry separates configurations by the dominant physical direction without using any $\mathbf{c}$ information in the forward pass; the Standard-Model-like

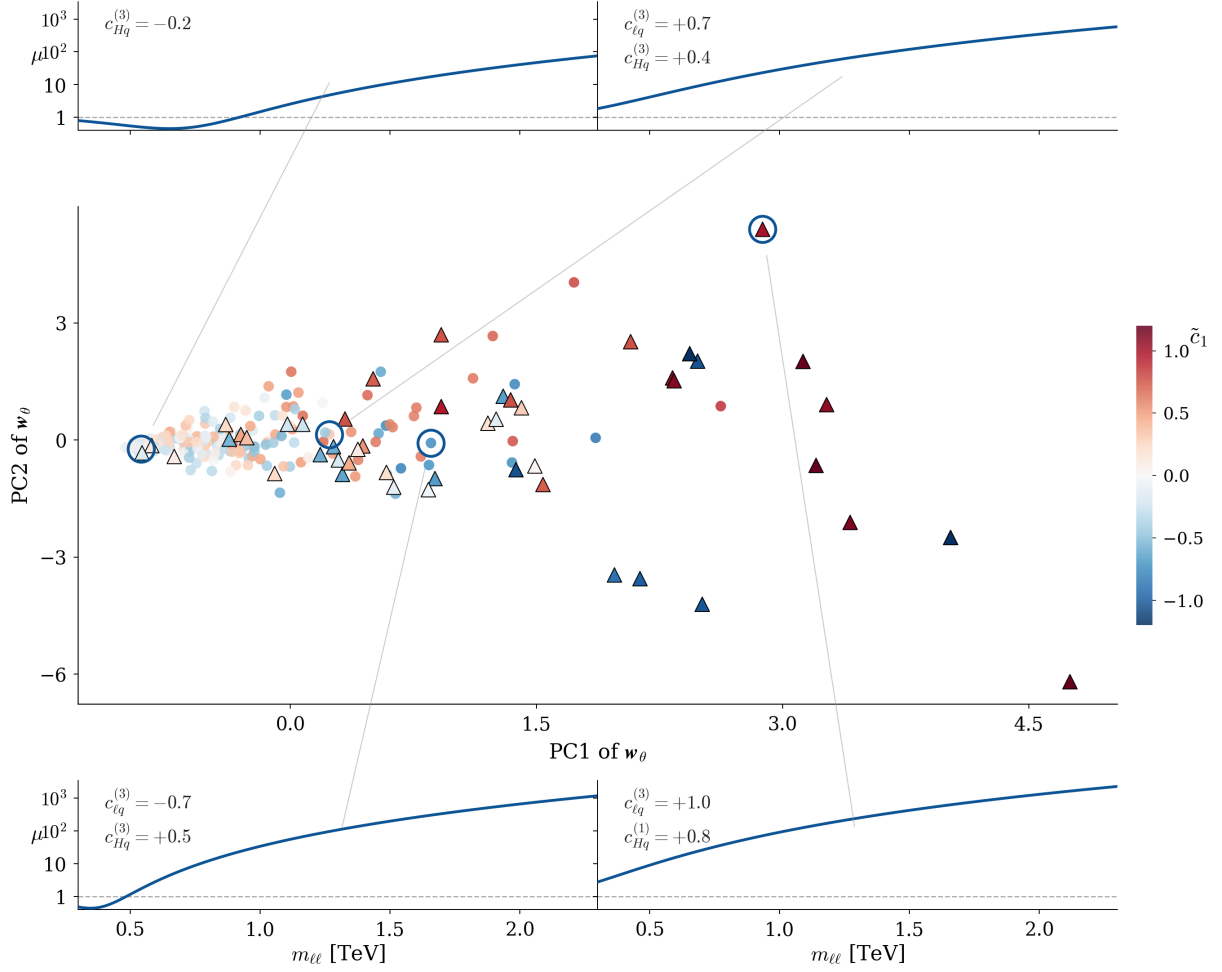


Figure 8: Ridge-weight latent geometry of the closed-form head over the SMEFT prior. Centre: the first two principal components of the implicit ridge weight $\mathbf{w}_\theta$ across 240 Wilson configurations. Circles are scenarios drawn from the training box $|c_i| \leq 0.7$; triangles are scenarios from the extrapolation annulus $0.7 < \max_i |c_i| \leq 1.0$. Colour encodes the leading Fisher-eigenbasis direction $\tilde{c}_1 \approx +0.64 c_{lq}^{(3)} - 0.77 c_{lq}^{(1)}$ from Section 5. Corners: four representative configurations, ringed in the scatter and linked by a thin grey connector to a thumbnail of the underlying spectrum. Each thumbnail plots $\mu(m_{\ell\ell}) = \sigma(\mathbf{c}, m_{\ell\ell}) / \sigma_{\text{SM}}(m_{\ell\ell})$ on a logarithmic vertical axis over $m_{\ell\ell} \in [0.3, 2.3]$ TeV, with the SM at $\mu = 1$ (dashed) and the two leading Wilson coefficients labelled in the upper-left corner.

configurations cluster around the centre, and extrapolation-annulus configurations populate the periphery.

The natural basis for reporting the Wilson posterior is the eigenbasis of the Fisher information of the oracle on the training Wilson prior, for reasons that have been independently arrived at by the SMEFT global-fit literature [33] and motivated in the broader uncertainty-quantification literature by the likelihood-informed subspace of Cui, Martin and Marzouk [34], the active-subspace construction of Constantine [35], and the optimal low-rank inverse-problem analysis of Spantini et al. [36].

We estimate the empirical Fisher

$$F_{ij} = \mathbb{E}_{\mathbf{c}, m_{\ell\ell}} [\partial_i \log \mu(\mathbf{c}, m_{\ell\ell}) \partial_j \log \mu(\mathbf{c}, m_{\ell\ell})] \quad (11)$$

on 600 Wilson configurations drawn from the same prior the foundation model is pretrained against, and eigendecompose $\mathbf{F} = \mathbf{V} \mathbf{D} \mathbf{V}^\top$ into descending eigenpairs. The eigenvalues are $\{19.6, 8.7, 3.0 \times 10^{-4}, 1.3 \times 10^{-4}\}$. The first two directions carry essentially all the kinematic

Table 3: Linear-probe readout R^2 on the analytic SMEFT oracle for the 5328-parameter learned-basis Intention head pretrained with the band $|c_{\ell q}^{(3)}| \in [0.6, 1.0]$ excluded. Top block: raw Warsaw basis. Bottom block: Fisher eigenbasis of the oracle on the training Wilson prior, with each rotated direction expressed as a linear combination of the Warsaw operators in the second column. Numbers from `experiments/full-chain-run/output/identifiability_summary.json`.

direction	inside-box R^2	withheld-band R^2
Warsaw basis		
$c_{Hq}^{(3)}$	-0.005	-0.227
$c_{Hq}^{(1)}$	-0.013	-0.090
$c_{\ell q}^{(3)}$	0.699	0.778
$c_{\ell q}^{(1)}$	0.146	-0.473
Fisher eigenbasis		
$\tilde{c}_1 \approx +0.64 c_{\ell q}^{(3)} - 0.77 c_{\ell q}^{(1)}$	0.686	0.735
$\tilde{c}_2 \approx -0.77 c_{\ell q}^{(3)} - 0.64 c_{\ell q}^{(1)}$	0.061	0.388
$\tilde{c}_3 \approx +0.76 c_{Hq}^{(3)} - 0.65 c_{Hq}^{(1)}$	0.004	-0.283
$\tilde{c}_4 \approx -0.65 c_{Hq}^{(3)} - 0.76 c_{Hq}^{(1)}$	-0.018	-0.040

information; the last two are degenerate at the 10^{-5} level with the available statistics. The columns of $\mathbf{V}$ define a rotated Wilson basis $\tilde{\mathbf{c}} = \mathbf{V}^\top \mathbf{c}$ in which the first two directions are linear combinations of the four-fermion operators $c_{\ell q}^{(3)}$ and $c_{\ell q}^{(1)}$ and the last two are linear combinations of the vertex operators $c_{Hq}^{(3)}$ and $c_{Hq}^{(1)}$. The rotation is a property of the observable and the Wilson prior, not of the foundation model.

For each held-out Wilson configuration $\mathbf{c}$ we draw a fresh context of $K = 12$ kinematic points from the analytic SMEFT oracle, evaluate the implicit ridge weight $\mathbf{w}_\theta(\mathbf{c})$, and fit a linear map $\tilde{\mathbf{c}} \approx \mathbf{W} \mathbf{w}_\theta + \mathbf{b}$ on the training-box scenarios. Table 3 reports per-direction coefficient of determination on a held-out in-box pool and on the withheld $|c_{\ell q}^{(3)}| \in [0.6, 1.0]$ band, both in the Warsaw basis and in the rotated eigenbasis. The asymmetry in the Warsaw basis has a physics interpretation: on the kinematic observable $m_{\ell\ell}$ alone, the two vertex operators produce rate-only modifications proportional to $(v/\Lambda)^2$ that are degenerate with the Standard-Model normalisation, so the linear probe cannot separate them from the SM offset; the two four-fermion operators produce identical $\log m_{\ell\ell}$ tails up to the relative weights of up-quark and down-quark luminosities, so the probe absorbs one combination and assigns the orthogonal combination the residual.

In the rotated basis the inside-box R^2 on the leading direction $\tilde{c}_1$ is 0.69 and the withheld-band R^2 is 0.74. The second direction reaches 0.06 inside the box and 0.39 in the withheld band; this in/out inversion is a pool-spread artefact rather than a recovery property, since the withheld annulus has roughly six-fold larger $\tilde{c}_2$ variance and the R^2 denominator SS_{tot} scales with it, so the operative diagnostic for $\tilde{c}_2$ is the per-direction MSE and coverage of Table 4 rather than R^2 . The two vertex-dominated directions sit at or below zero, consistent with the Fisher eigenvalues of 3×10^{-4} and 1.3×10^{-4} on those directions: they are not measurably resolved by the kinematic observable. We have not used the closed-loop run for this comparison: the linear probe is fit on 240 in-distribution scenarios drawn fresh per held-out evaluation, with the closed-form-head weights frozen from pretraining; the comparison is therefore a property of the closed-form head and the analytic oracle, not of the recovery dynamics.

5.1 Posterior coverage and the MLE floor

The point-estimate readout of Table 3 upgrades to a Bayesian posterior on $\tilde{\mathbf{c}}$ through the same Bayesian linear regression that defines the head. The linear map $\tilde{\mathbf{c}} \approx \mathbf{W} \mathbf{w}_\theta + \mathbf{b}$ propagates

Table 4: Per-direction Wilson posterior diagnostics in the Fisher eigenbasis for the 5 328-parameter learned-basis head pretrained with the band $|c_{\ell q}^{(3)}| \in [0.6, 1.0]$ excluded. Augmented linear-OLS probe with input $(\mathbf{w}_\theta, \log M_{\text{ctx}})$, against the bounded multi-restart MLE on the analytic likelihood with $K = 12$ contexts per scenario at $\sigma_y = 0.05$. $\text{MSE} = \text{bias}^2 + \text{variance}$ per scenario, averaged over the held-out pool with $N_{\text{rep}} = 100$ contexts per scenario. Numbers from `experiments/full-chain-run/output/bcrb_efficiency_augmented.json`.

direction	In-box pool				Withheld $ c_{\ell q}^{(3)} \in [0.6, 1.0]$			
	probe MSE	MLE MSE	ratio	cov 68%	probe MSE	MLE MSE	ratio	cov 68%
$\tilde{c}_1$ four-fermion (resolved)	0.035	0.108	0.33	0.88	0.092	0.441	0.21	0.82
$\tilde{c}_2$ four-fermion	0.165	0.132	1.24	0.48	0.256	0.125	2.05	0.48
$\tilde{c}_3$ vertex (prior-floored)	0.174	8.6	0.02	0.58	0.177	9.88	0.02	0.70
$\tilde{c}_4$ vertex (prior-floored)	0.186	7.4	0.03	0.52	0.244	9.53	0.03	0.58

the ridge-induced covariance $\sigma_y^2 \mathbf{A}^{-1}$ on $\mathbf{w}_\theta$ into a Gaussian covariance on $\tilde{\mathbf{c}}$. We additionally augment the probe input to the Bayesian-linear-regression sufficient pair $(\mathbf{w}_\theta, \log M_{\text{ctx}}) \in \mathbb{R}^{16+K}$ before fitting $\mathbf{W}, \mathbf{b}$ by ordinary least squares: the ridge weight $\mathbf{w}_\theta$ alone is a sufficient statistic for predicting μ at a new kinematic query, but the map back to $\tilde{\mathbf{c}}$ from $\mathbf{w}_\theta$ alone discards the conditioning on where in M the context sat, and restoring $\log M_{\text{ctx}}$ recovers the M -conditional sensitivity the maximum-likelihood estimator uses. The same OLS family stays inside the linear-readout framing. We report per-direction realised MSE on a held-out in-box pool of 50 scenarios and on the withheld $|c_{\ell q}^{(3)}| \in [0.6, 1.0]$ band, with $K = 12$ context observations per scenario, $N_{\text{rep}} = 100$ fresh contexts per held-out configuration, additive Gaussian noise on the cross-section ratio at $\sigma_y = 0.05$, and the probe matrices $\mathbf{W}, \mathbf{b}$ frozen from the pretraining-box fit. The reference benchmark is the per-context analytic maximum-likelihood estimator on the same contexts, evaluated with two random restarts inside a generous fit box $|c_i| \leq 5$.

The bounded MLE converges to an interior minimum on 72% of contexts (median gradient norm at the solution $\sim 10^{-4}$) and rests against the fit box on the remaining 28%, all in the $\tilde{c}_3, \tilde{c}_4$ vertex subspace where the $K = 12$ likelihood has shallow basins. The morphing of equation (2) is quadratic in $\mathbf{c}$ at fixed context, so the per-context likelihood admits multiple local minima from the sign ambiguity of $\sqrt{(\mu - \sigma_{\text{SM}})/A + (B/A)^2/4}$; the multi-restart MLE selects a basin per realization, which produces a finite- K sampling variance and bias that the asymptotic Cramér-Rao limit (efficient only in the linear regime) does not predict. The operational benchmark is therefore the realised MLE MSE at $K = 12$, $\sigma_y = 0.05$ rather than an asymptotic Fisher inverse. On $\tilde{c}_1$ the realised MLE has variance 0.063 and mean bias $|\bar{b}| = 0.11$ in-box (rising to variance 0.252 and $|\bar{b}| = 0.27$ on the withheld band); on $\tilde{c}_2$ the MLE variance is 0.058 and $|\bar{b}| = 0.11$ in-box. The augmented linear probe is a Bayesian shrinkage estimator with variance $v_{\text{probe}} = 0.009$ on $\tilde{c}_1$ in-box (0.040 withheld) and per-direction bias of comparable magnitude to the MLE.

On $\tilde{c}_1$ the probe dominates the MLE on both the variance term (a factor of seven in-box, a factor of six on the withheld band) and the bias term, so probe MSE is a factor of three below MLE MSE in-box (0.035 vs 0.108) and a factor of five below on the withheld band (0.092 vs 0.441). This is not a bias-for-variance trade: the probe wins outright on the most-resolved direction. On $\tilde{c}_2$ the probe variance is lower than the MLE (0.010 vs 0.058 in-box) but the prior shrinkage pulls the probe estimate further from the truth than the MLE pulls (probe $|\bar{b}| = 0.39$ vs MLE $|\bar{b}| = 0.27$), and the bias overwhelms the variance reduction; the MLE is the tighter estimator on $\tilde{c}_2$ at $1.24\times$ the MSE in-box and $2.05\times$ on the withheld band. This is the bias-for-variance trade going the wrong way for the probe, and it identifies $\tilde{c}_2$ as the direction the active-learning loop targets in Section 4.8. On the vertex directions the MLE is multimodal at this K and lacks a meaningful frequentist benchmark; the probe sits at the prior floor at $\text{MSE} \approx 0.18$, comparable to the prior variance, and dominates the MLE by a factor of 50.

$\tilde{c}_2$ **under-coverage and the active-learning target.** The $\tilde{c}_2$ probe MSE is dominated by per-context bias (typical $|\beta_a| \approx 0.39$ in-box, 0.47 withheld) rather than variance, and the 68% interval is correspondingly too narrow — 0.48 realised coverage against the nominal 0.68 on both pools. The four-fermion-only mass observable resolves $\tilde{c}_2$ at a Fisher eigenvalue of 8.7, but the per-context likelihood is too shallow for an unbiased estimator at $K = 12$ to reach a calibrated uncertainty, and the readout under-reports its uncertainty in the corresponding regime. The parameter-space EPIG of Section 4.8 targets this direction. The corrected angular observable of Section 2.1 contributes additional information on $\tilde{c}_2$ that the mass spectrum alone cannot saturate.

Disclosure-integrity gate. We additionally fit a parameter-matched MLP probe on the same $(\mathbf{w}_\theta, \log M_{\text{ctx}})$ features. If the closed-form-head representation carries the $\tilde{\mathbf{c}}$ structure, the linear and MLP probes should agree on the resolved directions on the withheld band; if the MLP pulls ahead OOD, work has migrated from the representation into the probe class. The linear-vs-MLP MSE ratio across the four directions on the withheld band reads 0.74, 0.72, 0.54, 0.81: the linear probe is at or below the MLP MSE on every direction, so the disclosure-integrity gate passes and the framing as a linear readout of the closed-form representation holds. The artifact $\mathbf{W}, \mathbf{b}$ and per-held-out-scenario posterior records are persisted at `experiments/full-chain-run/output/sbi_posterior_summary_mass_only.json`; the bounded-MLE-MSE summary at `bcrb_efficiency_augmented.json`.

6 Empirical comparison on the SMEFT oracle

The closed-form ridge solve in equation (3) has two architectural ingredients that can be evaluated separately: the closed-form solve itself, and the learned feature map ψ_θ . The principal comparison in this section is between the learned ψ_θ and a hand-engineered fixed polynomial basis under the same closed-form solve, which isolates what the learning contributes to the SMEFT regression. A second pair of comparisons positions the closed-form ridge against architectures that pool the context into a fixed-dimensional summary (Deep Sets) and against an architecture that consumes the Wilson vector at deployment (a constraint-violating regressor). The pooling comparison is an alignment ablation: it asks whether the joint $(M_{\text{ctx}}, Y_{\text{ctx}})$ alignment that the closed-form ridge exploits matters; the constraint-violating regressor sets the scale of what a multilayer perceptron with full $\mathbf{c}$ information at deployment can reach.

6.1 Setup

We use the Wilson-vector component order $\mathbf{c} = (c_{Hq}^{(3)}, c_{Hq}^{(1)}, c_{\ell q}^{(3)}, c_{\ell q}^{(1)})$ throughout. The training set is $N_{\text{train}} = 200$ scenarios with Wilson coefficients drawn uniformly from the hypercube $[-0.7, 0.7]^4$ over these four directions. Each scenario consists of a context of $K = 12$ kinematic points sampled uniformly in $\log(m_{\ell\ell}/M_{\text{ref}})$ over $m_{\ell\ell} \in [0.3, 2.3]$ TeV, with labels drawn from the analytic SMEFT oracle, and a query block of $Q = 32$ further kinematic points labelled identically. The held-out benchmark uses two disjoint scenario pools: 50 in-distribution scenarios drawn from the same hypercube and 50 out-of-distribution scenarios drawn by rejection sampling from the annulus $0.7 < \max_i |c_i| \leq 1.0$. We report median and fifth-percentile coefficient of determination (R^2) across the held-out pools.

6.2 Baselines

The first baseline is the fixed-basis Intention variant with $\psi(m) = \{1, \log(m/M_{\text{ref}}), \dots, \log^4(m/M_{\text{ref}})\}$, no trainable parameters, and the same closed-form ridge solve. This is the closed-form ridge in a hand-engineered polynomial basis and probes whether the learned basis ψ_θ does anything beyond rediscovering polynomial features. The second is a Deep Sets [4, 21] encoder that processes

each context observation through a per-event multilayer perceptron, mean-pools the resulting embeddings, concatenates the query, and decodes through a second multilayer perceptron at 6 545 parameters. The third is a constraint-violating multilayer-perceptron regressor that consumes the Wilson vector $\mathbf{c}$ directly on its forward pass, mapping $(\mathbf{c}, M_q) \rightarrow \hat{Y}_q$ at 5 041 parameters. It is not $\mathbf{c}$ -agnostic and so cannot participate in the closed-loop workflow of Section 4; we include it to set the scale of what predictive R^2 a multilayer perceptron can reach when it has the Wilson vector at deployment.

6.3 Learned versus fixed feature map

Table 5 reports the held-out comparison on the analytic SMEFT oracle. The principal contrast in the table is the first pair of rows: the fixed polynomial ψ reaches median $R^2 = 0.99$ in both regions but with fifth-percentile $R^2 = 0.91$ in-distribution and 0.76 out-of-distribution; the learned ψ_θ matches the median to machine precision ($R^2 = 0.9999$) and reduces the fifth-percentile gap to the median by roughly a factor of 20 in-distribution and a factor of 800 out-of-distribution. The fixed polynomial basis is therefore competitive on the median but brittle on tail scenarios where a low-order polynomial in $\log m_{\ell\ell}$ misses the four-fermion energy-growing tail; the learned basis is competitive on tail scenarios because ψ_θ acquires localised bumps and tail-emphasising shapes (see Section 6.4) that the polynomial cannot represent. The closed-form ridge solve does almost all of the inference work in both rows; the learned basis contributes the fifth-percentile stability.

The Deep Sets baseline and the constraint-violating regressor enter as alignment and information ablations. Deep Sets reaches median $R^2 = 0.70$ in-distribution and 0.51 out-of-distribution: a parameter-matched encoder that mean-pools per-event embeddings before decoding cannot recover the joint $(M_{\text{ctx}}, Y_{\text{ctx}})$ alignment that the closed-form ridge solve exploits by construction. The alignment ablation is the architectural point Deep Sets makes; the median number is not the point. The constraint-violating multilayer-perceptron regressor reaches median $R^2 = 0.96$ in both regions but has fifth-percentile $R^2 = -23$ in-distribution, consistent with a network that consumes $(\mathbf{c}, m)$ jointly memorising the training prior. The constraint-violating row sets the multilayer-perceptron-with-Wilson-vector ceiling on this regression problem; the closed-form ridge in the learned basis exceeds it without consuming $\mathbf{c}$.

6.4 The learned feature map

Figure 9 shows the 16 columns of ψ_θ trained on the SMEFT oracle alongside the 5 columns of the fixed polynomial basis. The learned columns include features with localised bumps near $m_{\ell\ell} \approx 0.4$ TeV and $m_{\ell\ell} \approx 1.5$ TeV, sigmoid-like transitions in $\log m$, and tail-emphasising shapes that the polynomial basis cannot represent. Several columns are visibly non-monotone. The fixed basis, by contrast, consists of increasing polynomials in $\log m$.

The learned basis produces a more numerically stable ridge problem on tail scenarios than the fixed basis does, at no cost to the median. This is the operational sense in which ψ_θ encodes structure from the SMEFT data manifold that the hand-engineered polynomial basis cannot. A comparison against a scaled-budget transformer Joint-Embedding Predictive Architecture is in Appendix C; the conclusion of that comparison is that JEPA-FM reaches comparable median R^2 at thirty times the parameter count and fifty times the training data, with linear-probe Wilson recoverability 0.62 against the closed-form head’s 0.75.

6.5 Sample efficiency

Figure 10 reports the held-out median R^2 as a function of the number of Adam steps. The learned-basis Intention head reaches $R^2 > 0.93$ after a single optimiser step on 24 scenarios, climbs to $R^2 > 0.97$ by step 25, and saturates at $R^2 = 0.9999$ around step 1 500. The Deep

Table 5: Held-out comparison on 50 in-distribution and 50 out-of-distribution SMEFT scenarios drawn from the analytic Drell-Yan oracle of Section 2. Median and fifth-percentile coefficient of determination (R^2) are reported, with mean-squared error on the query block. Parameter counts are matched to within $1.2\times$ across the trained architectures. The constraint-violating regressor consumes the Wilson vector on its forward pass and is not $\mathbf{c}$ -agnostic; we include it to set the scale of multilayer-perceptron performance with full $\mathbf{c}$ information at deployment. The MSE column is the unnormalised mean-squared error in μ -units and depends on the per-scenario scale of the cross-section ratio: in particular the in-distribution learned- ψ_θ MSE of 3142 is dominated by a single scenario at $R^2 = 0.32$ (index 44 of the held-out pool) where the seed context lands in a corner of the parameter space; the other 49 scenarios in that row sit at median $R^2 = 0.9999$. The out-of-distribution pool contains no comparable outlier, which is why its MSE is two orders of magnitude lower. The R^2 columns are scale-invariant and are the figure of merit for cross-architecture comparison. Numbers from `experiments/intention-vs-deepsets/output_smeft/summary.json`.

Architecture	Params	In-distribution			Out-of-distribution		
		median R^2	p5 R^2	MSE	median R^2	p5 R^2	MSE
Intention, fixed ψ	0	0.9936	0.914	5687	0.9884	0.756	10110
Intention, learned ψ_θ	5 328	0.9999	0.9996	3142	0.9999	0.9997	44
Deep Sets (matched)	6 545	0.703	-0.270	18696	0.511	0.190	80968
Constraint-violating regressor	5 041	0.960	-23.25	2156	0.960	-0.410	19063

Sets baseline sits at $R^2 \ll 0$ for the first ~ 200 steps, crosses zero around step 600, and reaches 0.70 only by step 1 500. The closed-form ridge solve does almost all of the inference work; the multilayer perceptron only needs to nudge ψ_θ into a basis in which the per-scenario ridge problem is well posed.

6.6 Foundation-model comparison on M - Y alignment retention

The drift-monitored inference loop uses four analytic properties of the closed-form head: the predictive variance, the closed-form information gain, the Sherman-Morrison sequential update, and the Eckart-Young condition-number sensitivity. The same loop on a foundation model that does not carry any of those four properties cannot work as a closed form. Table 6 reports the comparison across five architectures, ordered by their retention of M - Y alignment in the learned representation, measured as $1 - \cos$ between the embedding of a context ($M_{\text{ctx}}, Y_{\text{ctx}}$) and the embedding of the same context with the Y_{ctx} vector shuffled within the context. Higher alignment retention indicates that the representation uses the joint M - Y assignment rather than pooling it away. The four closed-form properties are determined by architectural type and are listed against each row.

The ordering on the four $\mathbf{c}$ -agnostic architectures is monotone in alignment retention. The fixed- ψ Intention variant retains less alignment than the learned- ψ_θ variant because the polynomial basis is constructed from m alone with no Y -conditional shape; the closed-form ridge solve restores most of the alignment information at inference but the static feature map already discards some. Scaled JEPA-FM at 186 241 parameters and 10 000 scenarios retains less alignment again, because the per-event encoder before pooling collapses the context into a fixed-dimensional summary that has to be decoded back into a Y -conditional prediction. Deep Sets pools the joint (M, Y) encoding to a fixed dimension before decoding and loses the alignment almost entirely. The $\mathbf{c}$ -recoverability column tracks the alignment-retention column on the same four rows: an architecture that pools the alignment away cannot recover Wilson directions from its representation, by the same mechanism that the closed-form ridge solve uses to read them out. The constraint-violating regressor consumes $\mathbf{c}$ on the forward pass, so the alignment-retention and $\mathbf{c}$ -recoverability axes do not apply to it; we include it as the held-out- R^2 ceiling that an

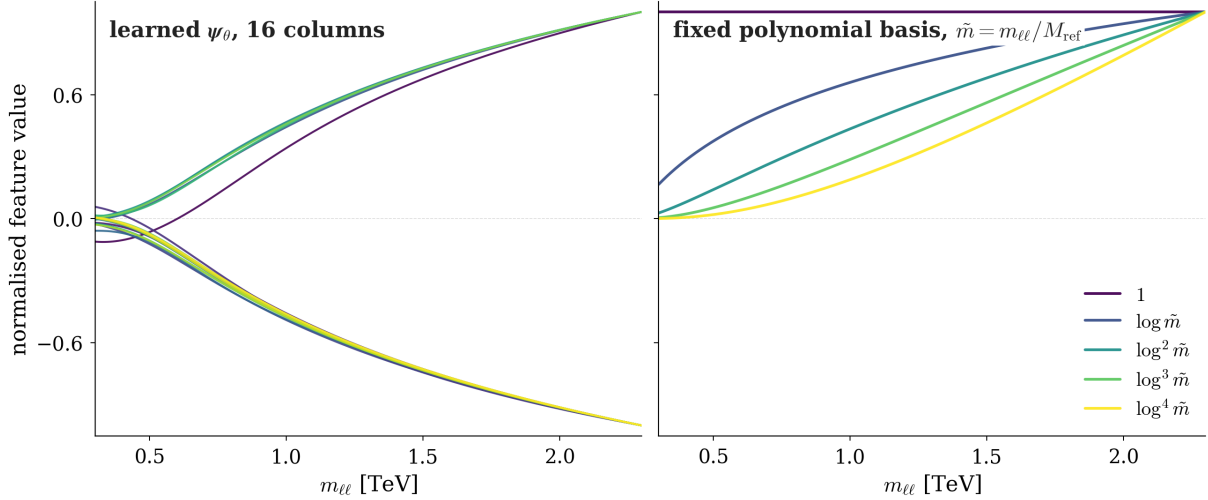


Figure 9: The learned feature map $\psi_\theta : \mathbb{R} \rightarrow \mathbb{R}^{16}$ on the SMEFT oracle, evaluated on a dense grid in $m_{\ell\ell}$ (left), and the fixed polynomial feature map $\psi(m) = \{1, \log(m/M_{\text{ref}}), \dots, \log^4(m/M_{\text{ref}})\}$ (right). The learned columns include localised bumps and tail-emphasising shapes that the fixed basis cannot represent.

Table 6: Foundation-model comparison on the SMEFT oracle, ordered descending by M - Y alignment retention. “Closed-form properties” lists whether the architecture admits the four analytic properties used by the drift-monitored inference loop: predictive variance (PV), information gain (IG), Sherman-Morrison update (SM), and Eckart-Young condition-number sensitivity (EY). Alignment retention is $1 - \cos$ of the embedding before and after shuffling Y_{ctx} within the context. Held-out median R^2 is on 50 out-of-distribution scenarios; C_{rec} is the joint R^2 of a linear probe onto the four Warsaw operators. The constraint-violating regressor consumes $\mathbf{c}$ on the forward pass; its alignment and $\mathbf{c}$ -recoverability cells are undefined by construction. Numbers from `experiments/full-chain-run/output/architecture_comparison.json`.

architecture	closed-form props	align. retained	median R^2_{out}	$C_{\text{rec,out}}$	params
Intention (closed-form, learned ψ_θ)	PV+IG+SM+EY	0.70	0.9999	0.65	5 328
Intention (closed-form, fixed ψ polynomial)	PV+IG+SM+EY	0.28	0.9884	0.66	0
Scaled JEPA-FM (transformer)	none	0.22	0.9695	0.51	186 241
Deep Sets matched-budget	none	0.03	0.5106	0.29	6 545
Constraint-violating regressor	none	N/A	0.9596	N/A	5 041

architecture with full $\mathbf{c}$ information at deployment reaches.

7 Discussion

We position the contribution against three lines of work and discuss the extension to measured data.

7.1 Foundation models in particle physics

The cluster of jet-level self-supervised foundation models [4–9] addresses a distinct task family from the present work. Those models are trained on large simulated jet corpora and evaluated on jet classification benchmarks, with success measured by area under the receiver-operating-characteristic curve and background rejection at fixed signal efficiency. The closest analogue in that cluster is the OmniLearn fine-tuning evaluation [6, 7], which reports sample-efficiency benefits on classification when the pretrained backbone is fine-tuned on small downstream labelled sets. We do not attempt a comparison to those benchmarks, because our model produces a

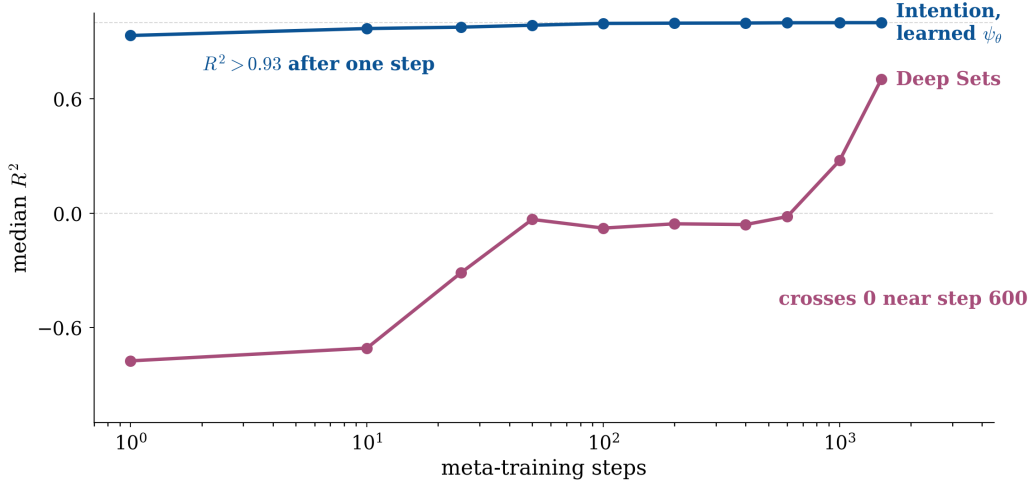


Figure 10: Held-out median R^2 on the in-distribution SMEFT scenarios as a function of the number of meta-training steps, logarithmic in the abscissa. The learned-basis Intention head reaches $R^2 > 0.93$ after one optimiser step and saturates near $R^2 = 0.9999$ by step 1 500. The Deep Sets baseline crosses $R^2 = 0$ near step 600 and plateaus at $R^2 = 0.70$.

continuous prediction of a cross-section ratio rather than a class probability; the underlying inference problem is different. We note one structural difference relevant to the present work: the jet-level cluster does not, in the published evaluations, equip the foundation model with closed-form predictive uncertainty, and the active-learning and conformal-calibration machinery of Section 4 therefore has no counterpart there.

The SMEFT-side foundation model of Das Bakshi, Hobbs and Kriesten [10], contemporary with the present work, is the closest neighbour. The two studies share the physics process and the dimension-six Warsaw operator set. They differ on three axes that bear on the inference problem. First, the Wilson-coefficient space: [10] sample 100 discrete SMEFT “universes” and train a supervised contrastive encoder to separate them in a 2-dimensional latent, while the present model is trained on a continuous prior over Wilson coefficients with the closed-form ridge head that is $\mathbf{c}$ -agnostic by construction. Second, the output: the contrastive encoder of [10] produces an embedding that is read out by an independently trained Dirichlet-prior-network classifier, anomaly-detection or nearest-neighbour head, while the present model produces a calibrated regression prediction directly from the closed-form ridge solve, with predictive uncertainty supplied by the same solve. Third, the inference workflow: [10] explicitly leave inference of Wilson coefficients from a measured distribution and the interplay with experimental data to future work, while the present model composes into the in-context inference loop of Section 4 and reads out a per-direction point estimate on the Fisher eigenbasis through the readout probe of Section 5, upgradeable to a credible interval under the same Bayesian linear regression.

7.2 SMEFT inference with machine learning

The closest physics-side antecedents are the parametrised-classifier line of Chen, Glioti, Marzocca, Nardini and Wulzer [30] and the joint-likelihood-ratio line of Brehmer, Cranmer, Louppe and Pavez [31] with the MadMiner software [32]. Both lines fit a network per fixed Wilson configuration sampled at training time and re-evaluate at deployment by interpolating across the trained Wilson grid. The present model amortises the inference rather than the conditional density: a single learned feature map handles arbitrary Wilson configurations through the context, with the only inference-time cost a matrix solve in a 16-dimensional space. The price is that the model is restricted to the observable representation in which ψ_θ is defined, here binned cross-section ratios in $m_{\ell\ell}$.

7.3 Prior-Fitted Networks and in-context inference

The Prior-Fitted Network construction [37, 38] is the direct methodological neighbour. In its modern form a transformer is pretrained on synthetic datasets drawn from a prior such that the forward pass through a context realises an approximate Bayesian posterior predictive; the recent TabPFN-as-SBI construction of Schmitt and Radev [38] explicitly recasts the head as a simulation-based-inference primitive. Our model is the closed-form-ridge specialisation of that family on a regression target whose inductive bias is exactly Bayesian ridge in a learned basis. The specialisation gives the four analytic properties of Section 1 that the generic transformer does not: closed-form predictive variance, closed-form information gain, closed-form sequential update, and closed-form Eckart-Young sensitivity. Without these the closed-loop monitoring of Section 4 would require numerical estimation of every monitoring quantity, which is not viable at acquisition-time cadences.

A parameter-matched transformer-based PFN trained on the same scenario prior for the same number of Adam steps reaches median $R^2 = 0.74$ in-distribution and 0.84 out-of-distribution on the polynomial-toy verification of Appendix B, well below the learned-basis Intention head at the same parameter budget. We read this as evidence that the closed-form ridge inductive bias is well-matched to the SMEFT regression problem at this parameter budget, rather than as a statement about the relative scaling of the two architectures.

7.4 Extension to measured cross-section ratios

The $\mathbf{c}$ -agnostic property of Section 3.2 licenses the substitution of a measured differential distribution for the analytic-oracle context: the closed-form head accepts any $(M_{\text{ctx}}, Y_{\text{ctx}})$ sequence with no provenance flag, and the loop machinery acts on whatever residual stream it receives. A demonstration of the substitution on a published high-mass Drell-Yan measurement, with the noise model relaxed to per-bin variances and the conformal calibrator using the experimental covariance matrix, is the immediate next step. The leading-order accuracy of the analytic oracle the foundation model is pretrained against is the operational constraint on this extension; a measured cross-section ratio absorbs higher-order QCD and electroweak corrections that the prior does not see, and these enter the loop as a systematic prior mismatch that the accuracy and calibration drift detectors will pick up but that no additional context can correct without a retrained ψ_θ .

7.5 Fisher-eigenbasis convention

The SMEFT global-fit literature has independently arrived at the Fisher eigenbasis as the right space in which to report sensitivity. The Top-Higgs-diboson-electroweak global fit of Ellis, Madigan, Mimasu, Sanz and You [33] reports sensitivities of the linear combination $\mathbf{V}^\top \mathbf{c}$ rather than the raw Warsaw $\mathbf{c}$ for the same physics reason that motivates Section 5: the kinematic observable resolves a small number of leading directions in Wilson space, and reporting in the unrotated basis either over-claims the readout of degenerate combinations or under-claims the readout of the one well-resolved direction. The present work imports this convention onto the foundation-model side; the readout probe in Section 5 is a linear map from the implicit ridge weight onto the rotated basis.

8 Summary and outlook

The research question asked whether drift monitoring can improve simulation-based inference in a physics foundation model through active learning. The current answer is a structured partial yes.

On the SBI component, the closed-form Bayesian linear regression in a learned feature map supplies a per-direction Wilson posterior on the Fisher eigenbasis of the oracle. The augmented linear probe attains MSE $0.33\times$ and $0.21\times$ the finite-sample bounded-MLE MSE on the leading resolved direction in-box and on the withheld band, dominating the MLE on both variance and bias; on the sub-leading direction the probe’s prior shrinkage adds more bias than it removes variance and the MLE is the tighter estimator, with under-coverage at the nominal 68% that the active-learning loop targets; the vertex directions are at the prior floor as their Fisher eigenvalues of $\sim 10^{-4}$ predict. The MLE confirms the prior-dominated floor on the vertex directions is the observable rather than the readout.

On the drift component, three detectors on standardised residuals, per-region conformal coverage, and the design-matrix condition number aggregate into an acquisition trigger. On the engineered $|c_{\ell q}^{(3)}| \in [0.6, 1.0]$ band the loop drives the target-band RMSE from 265 at seed context to 2.65 after 500 oracle calls, with no gradient-descent step at inference. The drift-triggered acquire-update-recalibrate sequence is the load-bearing mechanism, and its closure does not depend on which rule selects acquisition points within the drifted region.

On the active-learning component the present-tense finding is the contraction–inference divergence of Section 4.9. The 1D mass observable scored in cross-section-ratio root-mean-square error places the predictive target in a near-isotropic regime in which the Fisher spectrum’s five-orders-of-magnitude eigenvalue split is averaged away, and predictive EPIG of equation (7) does not outperform uniform random ($t = -2.82$, five-seed paired). The angular-observable extension breaks the four-fermion-vertex degeneracy and is validated against MadGraph at median $|\Delta A_{\text{FB}}| = 0.41\%$ across a 20-cell (c, m) grid at 64 000 events per cell, with one residual cell at $c_{\ell q}^{(1)} = 0.25$, $m_{\ell\ell} = 0.7$ TeV at 1.39% that localises a small remaining systematic in the up/down-luminosity weighting of the four-fermion singlet hadronic combination at moderate mass. The closed-form parameter-space EPIG of equation (10) is the corresponding $\tilde{c}$ -space acquisition score and reduces resolved-subspace differential entropy monotonically per the derivation. Across a 2 700-chain sweep on the angular observable at the stressed budget — a 1 800-chain design-only sweep across three pools, three K , two per-fire budgets, five acquisitions, and twenty seeds, plus a 900-chain closed-loop sweep that runs the per-cycle acquire-update-rescore loop out to $N_{\text{cycles}} = 500$ at $K \in \{12, 24, 48, 96\}$ — the parameter-space EPIG outperforms uniform random selection on $\Sigma_{\tilde{c}_2}$ contraction by 10–47 σ on every cell, and the analytic-morphing MLE on the same direction is worse than uniform random selection by 7–47 σ on every cell. The divergence sharpens with K and with cycle depth. The mechanism is that the top-decile information-gain candidates are perfectly collinear in the $\mathbf{A}^{-1}$ metric on every candidate pool tested, including the heterogeneous tail-spiked physical pool with coefficient-of-variation 1.10: targeted acquisition concentrates the posterior on a one-dimensional ridge of the learned feature space that does not contain the bimodal target’s identifying combination, while random selection spreads picks across the orthogonal $m_{\ell\ell}$ axis that does. The active-learning component therefore yields a calibrated negative: the closed-form contraction is at 30–40 σ a misleading proxy for inference quality on this physics. The closed-form ridge head supplies both the contraction and the MLE in $\mathcal{O}(D^2)$ per cycle, so the contraction-MLE gap is itself the online monitoring signal the loop should expose to its consumer, of the same arithmetic class as the four closed-form properties of Section 1.

The closed-form head, at 5 328 parameters, fits the analytic SMEFT oracle at median $R^2 = 0.9999$ on held-out scenarios. The architectural comparison of Section 6.6 orders the five candidate foundation models by their M – Y alignment retention; the four closed-form properties used by the drift loop and the linear-probe Wilson recoverability track that ordering monotonically across the c -agnostic architectures. The scaled JEPA-FM at 186 241 parameters and 10 000 scenarios reaches median $R^2 = 0.95$, characterising the parameter-and-data cost of learning the alignment instead of building it in.

Three extensions follow. The first is a batch-aware acquisition rule that the contraction-MLE diagnosis of Section 4.9 identifies as the next mechanism to test: with the top-decile-

information-gain candidates collinear in the $\mathbf{A}^{-1}$ metric, the greedy sequential-EPIG selector exhausts the one-dimensional ridge regardless of budget, and a joint-batch rule in the spirit of BatchBALD [20] that scores k -tuples instead of single picks is the candidate mechanism for breaking the redundancy. The second is the measured-data extension described in Section 7.4: consuming a published high-mass Drell-Yan differential measurement with its reported covariance matrix and a Mondrian-conformal calibrator on the experimental uncertainty, with the Fisher-eigenbasis readout compared against the published SMEFT bound on the same rotated directions. The third is the move from the one-dimensional binned-distribution observable used here to multi-dimensional event-level kinematics; the closed-form ridge solve is feature-map-agnostic, so a hybridisation in which a Particle Transformer [39] or Deep Sets event encoder produces the rows of the Intention head should preserve the four analytic properties, and the resulting feature space, of higher intrinsic dimension than the present scalar- $m_{\ell\ell}$ basis, is the natural setting in which the high-leverage subspace ceases to be one-dimensional and a targeted rule can plausibly separate from random on inference rather than on contraction alone.

Acknowledgements

The closed-form attention pattern is from Garnelo and Czarnecki [11]. The conformal-prediction substrate follows Araz and Spannowsky [17]. The SMEFT cross-section calculator follows Greljo and Marzocca [25] with Warsaw-basis operators from Grzadkowski, Iskrzynski, Misiak and Rosiek [24].

A Closed-form derivations

A.1 Sherman-Morrison sequential update

Given the inverse design matrix $\mathbf{A}^{-1}$ at context size K and a new context entry with feature vector $\psi_p = \psi_\theta(M_{K+1})$, the Sherman-Morrison identity [14] gives the rank-one update

$$\mathbf{A}_{\text{new}}^{-1} = \mathbf{A}^{-1} - \frac{(\mathbf{A}^{-1}\psi_p)(\psi_p^\top \mathbf{A}^{-1})}{1 + \psi_p^\top \mathbf{A}^{-1}\psi_p}, \quad (12)$$

which is exact to numerical precision. Equation (12) costs $\mathcal{O}(D^2)$ floating-point operations and admits a streaming implementation; the foundation model context can grow by one observation per Sherman-Morrison invocation, with the inverse design matrix maintained as a cache.

A consequence of equation (12) is the monotonicity of leverage. Let $\text{lev}_T = \psi_\theta(T)^\top \mathbf{A}^{-1}\psi_\theta(T)$ at context size K , and $\text{lev}_{T|p}$ at context size $K+1$ after the augmentation. Applying equation (12),

$$\text{lev}_{T|p} = \text{lev}_T - \frac{k_{Tp}^2}{1 + \text{lev}_p}, \quad (13)$$

with $k_{Tp} = \psi_\theta(T)^\top \mathbf{A}^{-1}\psi_\theta(p)$. The second term is non-negative, so $\text{lev}_{T|p} \leq \text{lev}_T$ for any target T and any augmentation p . The target-set predictive entropy $H_T = \frac{1}{2} \sum_i \log(2\pi e \sigma_y^2 \text{lev}_{T_i})$ is correspondingly monotone non-increasing under context growth.

A.2 Closed-form Expected Predictive Information Gain

The Expected Predictive Information Gain [13] on a target query T from a candidate pool point p , under the Bayesian linear regression on ψ_θ , is

$$\text{IG}_T(p) = \mathbb{E}_T \left[\frac{1}{2} \log \left(\frac{\sigma_T^2}{\sigma_{T|p}^2} \right) \right] = \frac{1}{2} \log \left(\frac{\text{lev}_T}{\text{lev}_{T|p}} \right) = \frac{1}{2} \log \left(\frac{\text{lev}_T}{\text{lev}_T - k_{Tp}^2 / (1 + \text{lev}_p)} \right), \quad (14)$$

where the second equality uses the homoscedastic-noise cancellation and the third uses equation (12). The Cauchy-Schwarz inequality applied to the $\mathbf{A}^{-1}$ inner product gives $k_{Tp}^2 = (\boldsymbol{\psi}(T)^\top \mathbf{A}^{-1} \boldsymbol{\psi}(p))^2 \leq (\boldsymbol{\psi}(T)^\top \mathbf{A}^{-1} \boldsymbol{\psi}(T))(\boldsymbol{\psi}(p)^\top \mathbf{A}^{-1} \boldsymbol{\psi}(p)) = \text{lev}_T \text{lev}_p$, which bounds the argument of the logarithm to $(0, 1]$ and renders $\text{IG}_T(p) \geq 0$.

A.3 Closed-form parameter-space EPIG

The Wilson posterior covariance in the resolved Fisher eigensubspace is $\boldsymbol{\Sigma} = \sigma_y^2 \mathbf{P} \mathbf{A}^{-1} \mathbf{P}^\top$ with $\mathbf{P} = \mathbf{V}^\top \mathbf{W} \in \mathbb{R}^{r \times D}$, $\mathbf{V}$ the Fisher rotation, $\mathbf{W}$ the frozen probe matrix of Section 5, and r the resolved-subspace dimension. The Sherman-Morrison identity of equation (12) propagates a candidate acquisition p with feature vector $\boldsymbol{\psi}_p = \boldsymbol{\psi}_\theta(M_p)$ to

$$\mathbf{A}_{\text{new}}^{-1} = \mathbf{A}^{-1} - \frac{(\mathbf{A}^{-1} \boldsymbol{\psi}_p)(\boldsymbol{\psi}_p^\top \mathbf{A}^{-1})}{1 + \text{lev}_p}. \quad (15)$$

Conjugating by $\mathbf{P}$ and writing $\mathbf{u} \equiv \mathbf{P} \mathbf{A}^{-1} \boldsymbol{\psi}_p \in \mathbb{R}^r$ gives the rank-one update

$$\boldsymbol{\Sigma}_{\text{new}} = \boldsymbol{\Sigma} - \sigma_y^2 \frac{\mathbf{u} \mathbf{u}^\top}{1 + \text{lev}_p}. \quad (16)$$

The matrix determinant lemma applied to the rank-one update gives the joint differential entropy reduction of the resolved subspace,

$$\Delta H_D(p) = \frac{1}{2} [\log \det \boldsymbol{\Sigma} - \log \det \boldsymbol{\Sigma}_{\text{new}}] = -\frac{1}{2} \log \left(1 - \frac{\sigma_y^2 \mathbf{u}^\top \boldsymbol{\Sigma}^{-1} \mathbf{u}}{1 + \text{lev}_p} \right), \quad (17)$$

and the single-direction posterior-variance reduction on $\tilde{c}_a$,

$$\Delta H_a(p) = -\frac{1}{2} \log \left(1 - \frac{\sigma_y^2 u_a^2}{(1 + \text{lev}_p) \Sigma_{aa}} \right). \quad (18)$$

The quadratic form $\mathbf{u}^\top \boldsymbol{\Sigma}^{-1} \mathbf{u} \geq 0$ by positive-definiteness of $\boldsymbol{\Sigma}^{-1}$; the Cauchy-Schwarz inequality applied to the $\boldsymbol{\Sigma}^{-1}$ inner product gives $u_a^2 \leq \Sigma_{aa}^{-1} \Sigma_{aa} u_a^2$ which keeps the log argument in $(0, 1]$, so both scores are non-negative. The computation reuses the existing Sherman-Morrison cache; the cost per candidate is $\mathcal{O}(rD + r^2)$ from the matrix-vector product and the resolved-subspace inverse, both of which are cached across candidates. The resolved-subspace dimension r is fixed to the data-dominated eigendirections identified in Section 5; including a near-degenerate direction in $\boldsymbol{\Sigma}^{-1}$ would cause $\det \boldsymbol{\Sigma}$ to underflow and the score to mis-rank candidates, which the eigenvalue gap on the SMEFT oracle makes a sharp partition.

A.4 Aggregator decision rules

The aggregator of Section 4.2 combines the three signal outputs through an eight-row table indexed by the binary states of accuracy drift, calibration drift, and the $\mathbf{v}_{\min}$ projection-ratio context-shape signal. When only calibration drift fires, the action is recalibration of the conformal calibrator. When accuracy drift fires alone or in combination, the action is local retraining inside the drifted region by acquiring additional context observations there. When all three fire, the action is global retraining (acquisition across the full kinematic axis). A calibration-only fire that recurs for four consecutive cycles escalates to local retraining; an accuracy-only fire requires three consecutive cycles before triggering acquisition; a cool-down of three cycles between consecutive local-retraining firings prevents back-to-back acquisitions on the same drift event.

B Polynomial-toy verification

The main comparison of Section 6 is on the analytic SMEFT oracle of Section 2. To verify that the closed-form ridge solve recovers the SMEFT morphing structure of equation (2) when the structure is exactly polynomial, we ran an additional comparison on a controlled polynomial-toy oracle.

The polynomial-toy oracle samples $\mu(\mathbf{c}, m_{\ell\ell}) = \phi_{\text{joint}}(\mathbf{c}, m_{\ell\ell}) \cdot \mathbf{W}$ with ϕ_{joint} the joint polynomial in $\mathbf{c}$ and $\log(m_{\ell\ell}/M_{\text{ref}})$ and $\mathbf{W}$ a fixed random ground-truth vector. This is the limit in which the morphing decomposition of equation (2) is exact at fixed $\mathbf{c}$. The fixed-basis Intention variant on the polynomial-toy oracle has the exact analytic structure of the target written into its feature map and reaches median $R^2 = 0.99994$ on both held-out pools, by construction. The learned-basis variant reaches median $R^2 = 0.99973$ in-distribution and 0.99962 out-of-distribution, with fifth-percentile $R^2 = 0.964$ and 0.927. A parameter-matched Deep Sets baseline reaches median $R^2 = 0.617$ in-distribution and 0.763 out-of-distribution. A parameter-matched Prior-Fitted Network reaches median $R^2 = 0.738$ and 0.839. The constraint-violating regressor reaches median $R^2 = 0.938$ and 0.935. The qualitative ordering matches Table 5; the absolute numbers are higher on the polynomial-toy oracle because the kinematic structure is exactly representable by the closed-form ridge solve in either basis.

The polynomial-toy comparison serves as a sanity check that the architecture and training procedure recover the morphing structure when the structure is present. The Section 6 comparison on the analytic SMEFT oracle is the result we read as evidence about the SMEFT inference problem.

C Representation-level diagnostics

A complementary set of diagnostics asks what each architecture’s representation encodes about the SMEFT structure of the data. We compare the closed-form Intention head against a scaled-budget transformer Joint-Embedding Predictive Architecture (JEPA-FM) implemented following the T-JEPA recipe of Thimonier et al. [23]: a per-event encoder $(\log m_i, Y_i) \rightarrow \mathbb{R}^{16}$ shared between context and exponential-moving-average target branches, a self-attention aggregator over the context with a learnable [CLS] token, a predictor that consumes a query-position encoding, and an auxiliary scalar decoder reading out $\hat{Y}_q$. The training loss is $\mathcal{L} = \mathcal{L}_{\text{JEPA}} + 0.04 \mathcal{L}_{\text{VICReg}} + 0.1 \mathcal{L}_{\text{MSE}}$ with the variance-covariance collapse regulariser of Bardes, Ponce and LeCun [40]. At 186 241 parameters trained on 10 000 scenarios for 10 000 steps, JEPA-FM reaches median $R^2 = 0.95$ in-distribution and 0.97 out-of-distribution, comparable to the closed-form Intention head at thirty times the parameter count and fifty times the training data. A linear probe from each representation onto the four Wilson coefficients fits with $R^2 = 0.75$ on the Intention ridge weight (in-distribution) against 0.62 on the scaled-budget JEPA-FM; the effective rank (singular values above 1% of the maximum) is 11 for Intention against 21 for JEPA-FM, with the first two-to-three directions carrying 90% of the variance in both representations consistent with the Fisher eigenvalues of $\{19.6, 8.7, 3.0 \times 10^{-4}, 1.3 \times 10^{-4}\}$ from Section 5. The mechanism in one clause: the closed-form ridge solve is by construction a function of the joint M – Y assignment through the right-hand product of equation (3), so any pooling or attention aggregator must spend capacity learning the alignment the closed form has built in, and that is the parameter and data gap.

D Reproduction

All experiments reported in this paper run on a single central-processing-unit. The closed-form attention head and the meta-learning objective are implemented in PyTorch through the differentiable solver `torch.linalg.solve`. The analytic oracle is implemented in NumPy with

parton distribution functions supplied by LHAPDF [27] using the CT18NNLO set [26]. The closed-loop run, including drift detection, EPIG acquisition and conformal recalibration, completes in ~ 17 minutes of pretraining and a further 19 seconds of loop time. The full source for the experiments and the closed-loop run is in the project repository at `experiments/intention-vs-deepsets/` and `experiments/full-chain-run/`; the numerical content of each table and figure can be regenerated from the JSON summaries cited in the corresponding captions. The plotting scripts that produce every figure in the paper, including a cycle-by-cycle animation of the prediction at `figures/learning_loop.gif`, are at `paper/scripts/`.

References

- [1] K. Cranmer, J. Brehmer, and G. Louppe. The frontier of simulation-based inference. *Proc. Natl. Acad. Sci.*, 117:30055, 2020.
- [2] A. Tejero-Cantero, J. Boelts, M. Deistler, J.-M. Lueckmann, C. Durkan, P. J. Goncalves, D. S. Greenberg, and J. H. Macke. sbi: a toolkit for simulation-based inference. *J. Open Source Software*, 5:2505, 2020.
- [3] J. Hermans, V. Begy, and G. Louppe. Likelihood-free MCMC with amortized approximate likelihood ratios, 2020.
- [4] L. Heinrich, T. Golling, G. Kasieczka, and et al. Masked particle modelling, 2024.
- [5] J. Birk, A. Hallin, and G. Kasieczka. OmniJet- α : The first cross-task foundation model for particle physics, 2024.
- [6] V. Mikuni and B. Nachman. OmniLearn: A method to simultaneously facilitate all jet physics tasks, 2024.
- [7] V. Mikuni and B. Nachman. OmniLearned: A foundation model for jet physics at the billion-jet scale, 2025.
- [8] J. Bardhan, R. Agrawal, A. Tilak, C. Neeraj, and S. Mitra. HEP-JEPA: A foundation model for collider physics using joint embedding predictive architecture, 2025.
- [9] Sophon Collaboration. Sophon: Foundation model for collider physics, 2024.
- [10] S. Das Bakshi, T. J. Hobbs, and B. Kriesten. Reusable theory representations for colliders: a demonstrator SMEFT foundation model, 2025.
- [11] M. Garnelo and W. M. Czarnecki. Exploring the Space of Key-Value-Query Models with Intention, 2023.
- [12] C. E. Rasmussen and C. K. I. Williams. *Gaussian processes for machine learning*. MIT Press, 2006.
- [13] F. Smith, B. Bickford Smith, and T. Rainforth. Prediction-oriented Bayesian active learning, 2023.
- [14] J. Sherman and W. J. Morrison. Adjustment of an inverse matrix corresponding to a change in one element of a given matrix. *Ann. Math. Statist.*, 21:124, 1950.
- [15] J. Lei, M. G’Sell, A. Rinaldo, R. J. Tibshirani, and L. Wasserman. Distribution-free predictive inference for regression. *J. Amer. Statist. Assoc.*, 113:1094, 2018.
- [16] V. Vovk, A. Gammerman, and G. Shafer. *Algorithmic learning in a random world*. Springer, 2005.

- [17] J. Y. Araz and M. Spannowsky. Another fit bites the dust: conformal prediction for calibrated uncertainty in high-energy physics, 2025.
- [18] Y. Benjamini and Y. Hochberg. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *J. Roy. Statist. Soc. B*, 57:289, 1995.
- [19] N. Ahad, M. A. Davenport, and Y. Xie. Data-adaptive symmetric CUSUM for sequential change detection, 2022.
- [20] A. Kirsch, J. van Amersfoort, and Y. Gal. BatchBALD: efficient and diverse batch acquisition for deep Bayesian active learning, 2019.
- [21] M. Zaheer, S. Kottur, S. Ravanbakhsh, B. Póczos, R. Salakhutdinov, and A. J. Smola. Deep sets. In *NeurIPS*, 2017.
- [22] M. Assran, Q. Duval, I. Misra, P. Bojanowski, P. Vincent, M. Rabbat, Y. LeCun, and N. Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *CVPR*, 2023.
- [23] H. Thimonier, J. De Melo Costa, F. Popineau, A. Rimmel, and B.-L. Doan. T-JEPA: augmentation-free self-supervised learning for tabular data. In *ICLR*, 2025.
- [24] B. Grzadkowski, M. Iskrzynski, M. Misiak, and J. Rosiek. Dimension-six terms in the Standard Model Lagrangian. *JHEP*, 10:085, 2010.
- [25] A. Greljo and D. Marzocca. High- p_T dilepton tails and flavour physics. *Eur. Phys. J. C*, 77: 548, 2017.
- [26] T.-J. Hou and et al. New CTEQ global analysis of quantum chromodynamics with high-precision data from the LHC. *Phys. Rev. D*, 103:014013, 2021.
- [27] A. Buckley, J. Ferrando, S. Lloyd, K. Nordström, B. Page, M. Rüfenacht, M. Schönherr, and G. Watt. LHAPDF6: parton density access in the LHC precision era. *Eur. Phys. J. C*, 75:132, 2015.
- [28] J. Alwall and et al. The automated computation of tree-level and next-to-leading order differential cross sections, and their matching to parton shower simulations. *JHEP*, 07:079, 2014.
- [29] R. Boughezal, Y. Huang, and F. Petriello. Impact of high invariant-mass drell-yan forward-backward asymmetry measurements on SMEFT fits, 2023.
- [30] S. Chen, A. Glioti, D. Marzocca, G. Nardini, and A. Wulzer. Unbinned multivariate observables for global SMEFT analyses from machine learning. *JHEP*, 03:033, 2023.
- [31] J. Brehmer, K. Cranmer, G. Louppe, and J. Pavez. Mining gold from implicit models to improve likelihood-free inference. *Proc. Nat. Acad. Sci.*, 117:5242, 2020.
- [32] J. Brehmer, F. Kling, I. Espejo, and K. Cranmer. MadMiner: Machine learning-based inference for particle physics. *Comput. Softw. Big Sci.*, 4:3, 2020.
- [33] J. Ellis, M. Madigan, K. Mimasu, V. Sanz, and T. You. Top, Higgs, diboson and electroweak fit to the Standard Model effective field theory, 2020.
- [34] T. Cui, J. Martin, Y. M. Marzouk, A. Solonen, and A. Spantini. Likelihood-informed dimension reduction for nonlinear inverse problems. *Inverse Problems*, 30:114015, 2014.

- [35] P. G. Constantine. *Active Subspaces: Emerging Ideas for Dimension Reduction in Parameter Studies*. SIAM, 2015.
- [36] A. Spantini, A. Solonen, T. Cui, J. Martin, L. Tenorio, and Y. M. Marzouk. Optimal low-rank approximations of Bayesian linear inverse problems. *SIAM J. Sci. Comput.*, 37: A2451–A2487, 2015.
- [37] S. Müller, N. Hollmann, S. P. Arango, J. Grabocka, and F. Hutter. Transformers can do Bayesian inference, 2022.
- [38] M. Schmitt and S. T. Radev. TabPFN-as-SBI: simulation-based inference with prior-fitted networks, 2025.
- [39] H. Qu, C. Li, and S. Qian. Particle Transformer for jet tagging. *Proc. 39th ICML*, 2022.
- [40] A. Bardes, J. Ponce, and Y. LeCun. VICReg: variance-invariance-covariance regularization for self-supervised learning. In *ICLR*, 2022.